Review

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Baveno VIII – Advancing consensus in portal hypertension ☆

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Summary

Since the first Baveno Consensus Conference in 1990, successive conferences held every 5 years have transformed the field of cirrhosis and portal hypertension, evolving from a focus on variceal bleeding to a pathophysiology-based framework centred on risk stratification, clinically significant portal hypertension (CSPH), and the prevention of decompensation. Building on this progress, the Baveno Cooperation, in collaboration with the Vascular Liver Disease Interest Group (VALDIG), convened an expert faculty to review the evidence published since Baveno VII and update clinical guidance for patients with advanced chronic liver disease (ACLD), portal hypertension, and vascular liver diseases. The Baveno VIII Consensus Conference, held in March 2026 in Baveno, Italy, was based on a structured consensus process across nine panels addressing key domains of ACLD, portal hypertension, and vascular liver diseases. Statements and recommendations were developed through iterative review of the evidence and voting among faculty, with strong consensus defined as >80% faculty agreement. A total of 272 statements and recommendations achieved strong consensus. Key areas included non-invasive risk stratification in compensated ACLD (cACLD), modifiers of steatotic liver disease, prevention of first and further decompensation, management of portal hypertensive bleeding and further decompensation, cirrhosis recompensation, and vascular liver diseases, including Budd–Chiari syndrome, portosinusoidal vascular disorder (PSVD) or non-cirrhotic portal fibrosis (NCPF), and portal vein thrombosis (PVT). The central role of CSPH and the expanding use of non-invasive tests were reaffirmed, and several diagnostic and prognostic cut-offs were revised. New to Baveno VIII is the inclusion of selected guidance for paediatric patients. Recommendations from previous conferences that were not specifically revised in Baveno VIII, such as those on hepatic venous pressure gradient (HVPG) measurement, remain valid. Finally, a research agenda identifying 153 priority topics is proposed.

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Introduction

The Baveno Cooperation, an official consortium of the European Association for the Study of the Liver (EASL) established in its current form in 2017, aims to advance knowledge in advanced chronic liver disease (ACLD), vascular liver diseases and portal hypertension. Beginning with the first meeting, organised in Groningen, the Netherlands, in 1986, subsequent consensus workshops have aimed to harmonise definitions, guide clinical research, and develop consensus-based recommendations, contributing to a shift towards a stage-based, prevention-oriented approach to ACLD/cirrhosis and portal hypertension. The academic rigour and clinical relevance of these initiatives have led to the organisation of a series of global consensus meetings: Baveno I (1990, Baveno, Italy); Baveno II (1995, Milan, Italy); Baveno III (2000, Stresa, Italy); Baveno IV (2005, Baveno, Italy); Baveno V (2010, Stresa, Italy); Baveno VI (2015, Baveno, Italy); Baveno VII (2021, virtual conference due to the COVID-19 pandemic); and finally, Baveno VIII (2026, Baveno, Italy).[1–7]

The Baveno VIII Consensus Conference (2026) provides updated statements and recommendations based on evidence accrued since Baveno VII. The statements and recommendations are presented across nine panels: (1) non-invasive test (NIT)-based risk stratification in compensated ACLD (cACLD); (2) modifiers of steatotic liver disease (SLD); (3) prevention of first decompensation; (4) prevention of further decompensation; (5) management of portal hypertension–related gastrointestinal bleeding; (6) management of further decompensation; (7) cirrhosis recompensation; (8) vascular liver diseases, including Budd–Chiari syndrome (BCS), and portosinusoidal vascular disorder (PSVD) also termed non-cirrhotic portal fibrosis (NCPF); and (9) portal vein thrombosis (PVT) both in patients without underlying liver disease and in patients with ACLD. For the first time, paediatric hepatologists were included in some of the panels and the document includes statements related to management of paediatric liver disease and portal hypertension.

☆ Endorsed by the European Association of the Study of the Liver (EASL), the European Society of Gastrointestinal Endoscopy (ESGE), and the European Reference Network for rare Hepatological Diseases (ERN RARE-LIVER).

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Each panel was composed of two chairs and 4–8 panellists (n = 79). Proposed statements were iteratively reviewed by the full Baveno faculty (n = 84), including chairs, panellists, and invited lecturers (n = 5) and were subjected to consensus voting. Those achieving ≥80% agreement, were approved and are presented below, organized by panel. In total, 272 statements and recommendations achieved consensus, all of which were graded according to level of evidence (according to the Oxford Centre for Evidence-Based Medicine; Table 1) and classified as new, modified or unchanged compared with Baveno VII; for recommendations, strength of recommendation was also reported (Table 2). Each panel also provided a research agenda including a total of 153 priority topics. Finally, a selection of statements and recommendations of previous Baveno consensus reports that remain valid but were not specifically updated for Baveno VIII are shown in Table 3.

Panel 1 – Risk stratification in cACLD

1.1. The term "compensated advanced chronic liver disease (cACLD)" reflects the continuum of severe fibrosis and cirrhosis. A pragmatic definition of cACLD based on liver stiffness measurement (LSM)[*0] aims to identify those at risk of CSPH, decompensation, and liver-related death at point of care, irrespective of histological stage or the ability of LSM to identify these stages **(LoE 3; modified; 100% agreement).**

1.2. "Compensated cirrhosis" and "cACLD" are not interchangeable, and the latter is preferred if the diagnosis is established by non-invasive tests (NITs) **(LoE 3; modified; 97% agreement).**

1.3. LSM values <10 kPa in the absence of other known clinical/imaging signs should be used to rule out cACLD; values between 10 and 15 kPa should be considered suggestive of cACLD; values >15 kPa should be considered highly suggestive of cACLD (Fig. 1) **(LoE 3; strong recommendation; modified; 100% agreement).**

1.4. LSM should always (*i.e.* for diagnosis of cACLD or CSPH) be performed under fasting conditions. In patients with an index LSM ≥10 kPa, LSM should either be repeated **(LoE 2; strong recommendation)**, or may be complemented by another validated NIT of advanced fibrosis **(LoE 3; weak recommendation)** to confirm cACLD **(modified; 97% agreement).**

1.5. Patients with chronic liver disease (which does not include vascular liver disease – see respective section) and an LSM <10 kPa have a negligible 3-year risk (≤1%) of decompensation and liver-related death **(LoE 1; modified – minimal; 98% agreement).**

1.6. LSM values <10 kPa in patients with clinical/imaging signs of portal hypertension should raise the suspicion for vascular liver disease (see respective section) **(LoE 3; strong recommendation; new; 100% agreement).**

1.7. Patients with cACLD should be referred to a liver disease specialist for further work-up **(LoE 3; strong recommendation; unchanged; 100% agreement).**

1.8. In patients with cACLD, LSM holds prognostic information **(LoE 1).** A rule of 5 for LSM (10-15-20-25 kPa) denotes progressively higher relative risks of decompensation and liver-related death, independently of the aetiology of chronic liver disease **(LoE 2; modified; 100% agreement).**

1.9. In patients with cACLD, NITs for CSPH (see below) should be repeated every 12 months **(LoE 3; strong recommendation; modified; 97% agreement).**

1.10. Changes in LSM confer prognostic information; however, the most recent value should guide clinical decision-making **(LoE 2; strong recommendation; new; 97% agreement).**

1.11. Although the concept of CSPH is based on hepatic venous pressure gradient (HVPG), validated elastography and validated NITs (see 1.12-1.16) are sufficiently accurate to exclude or identify CSPH and are similarly predictive of hepatic decompensation in cACLD **(LoE 1).** The body of evidence for rare aetiologies is more limited **(modified; 97% agreement).**

1.12. LSM and platelet count (PLT), as well as their derivatives (*i.e.* ANTICIPATE models) are the mainstay of non-invasive diagnosis of CSPH **(LoE 1; modified; 94% agreement).**

1.13. Spleen stiffness measurement (SSM) may further improve the classification of CSPH **(LoE 2; new; 94% agreement).**

1.14. ANTICIPATE (ANTICIPATE-NASH for people with metabolic dysfunction-associated steatotic liver disease [MASLD] and obesity [≥30 kg/m^2]) **(LoE 1; strong recommendation)** or NICER **(LoE 2; strong recommendation)** should be used to non-invasively estimate the probability of CSPH in cACLD **(new; 97% agreement).**

Table 1. Level of evidence based on the Oxford Centre for Evidence-based Medicine.

Level	Criteria	Simple model for high, intermediate and low evidence
1	Systematic reviews (SR) (with homogeneity) of randomised-controlled trials (RCT)	Further research is unlikely to change our confidence in the estimate of benefit and risk
2	RCT or observational studies with dramatic effects; SR of lower quality studies (*i.e.* non-randomised, retrospective)	
3	Non-randomised-controlled cohort/follow-up study/control arm of randomised trial (systematic review is generally better than an individual study)	Further research (if performed) is likely to have an impact on our confidence in the estimate of benefit and risk and may change the estimate
4	Case-series, case-control, or historically controlled studies (systematic review is generally better than an individual study)	
5	Expert opinion (mechanism-based reasoning)	Any estimate of effect is uncertain

Table 2. Grades of recommendation.

Grade	Wording	Criteria
Strong	Shall, should, is recommended. Shall not, should not, is not recommended	Evidence, consistency of studies, risk-benefit ratio, patient preferences, ethical obligations, feasibility
Weak or open	Can, may, is suggested. May not, is not suggested	

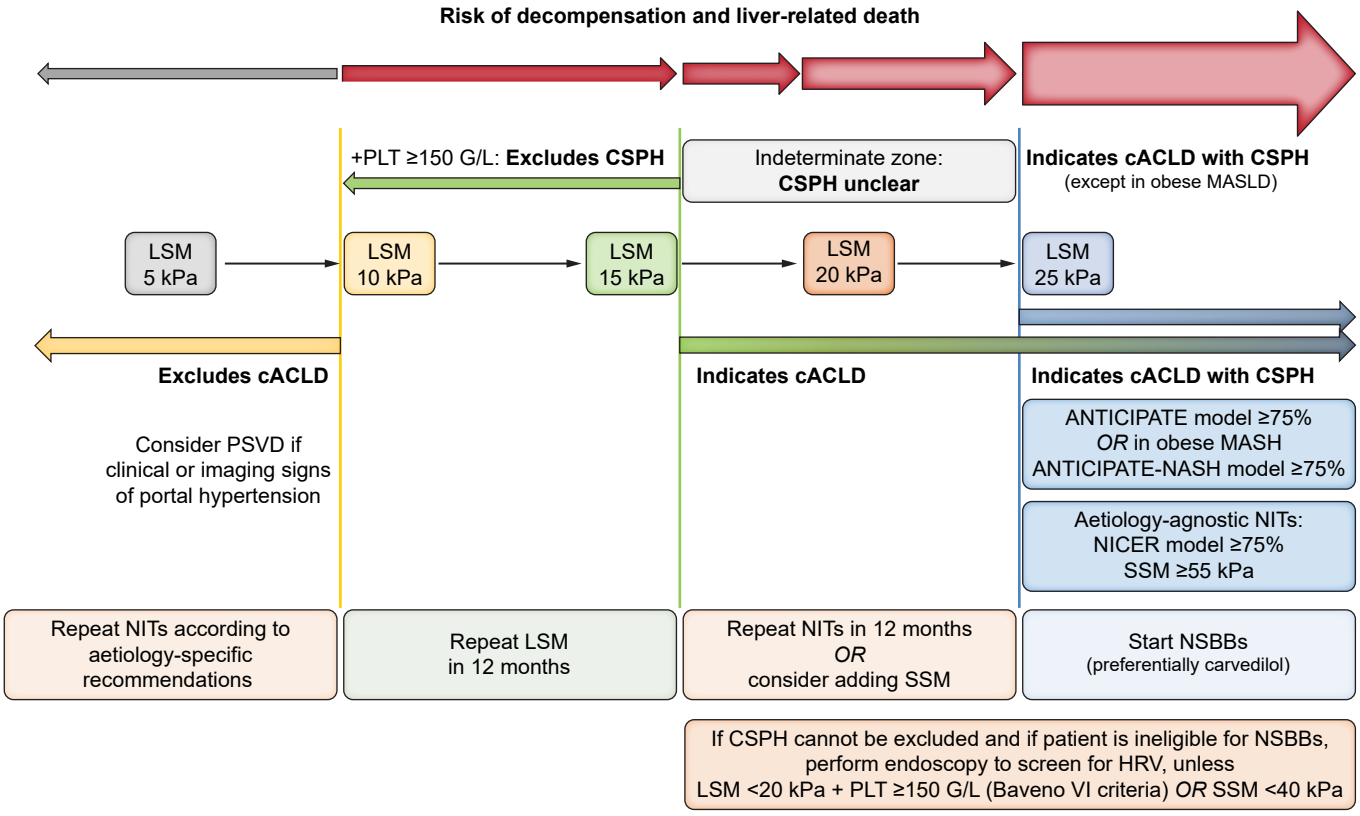


Fig. 1. Non-invasive diagnosis of cACLD and CSPH. cACLD, compensated advanced chronic liver disease; CSPH, clinically significant portal hypertension; HRV, high-risk varices; LSM, liver stiffness measurement; NITs, non-invasive tests; MASH, metabolic dysfunction-associated steatohepatitis; NSBBs, non-selective beta blockers; PLT, platelet count; PSVD, portosinusoidal vascular disorder; SSM, spleen stiffness measurement.

1.15. In patients with cACLD, LSM ≤15 kPa and PLT ≥150×10^9/L rule out CSPH (negative predictive value >90%) **(LoE 1; strong recommendation; modified; 100% agreement).**

1.16. In patients with cACLD, the following criteria are recommended to establish the diagnosis of CSPH, while considering their individual limitations **(new)**:

a. Estimated probability of CSPH ≥75% by ANTICIPATE/ANTICIPATE-NASH **(LoE 1; strong recommendation)** or NICER **(LoE 2; strong recommendation; 93% agreement).**

b. In virus- and/or alcohol-related cACLD and non-obese (BMI <30 kg/m^2) MASLD-related cACLD, LSM ≥25 kPa **(LoE 1; strong recommendation; 91% agreement).**

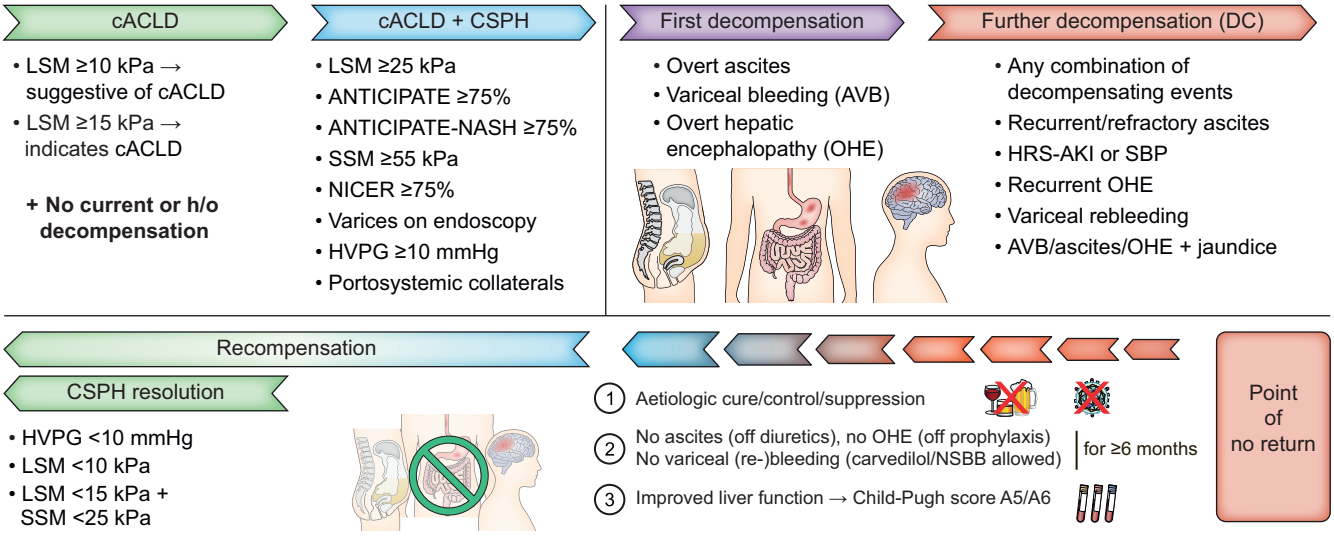


Fig. 2. Definitions of the different states of cirrhosis/ACLD. AVB, acute variceal bleeding; cACLD, compensated advanced chronic liver disease; CSPH, clinically significant portal hypertension; (O)HE, (overt) hepatic encephalopathy; HRS-AKI, hepatorenal syndrome-acute kidney injury; HVPG, hepatic venous pressure gradient; LSM, liver stiffness measurement; SBP, spontaneous bacterial peritonitis; SSM, spleen stiffness measurement.

c. SSM (100 Hz) >55 kPa **(LoE 2; strong recommendation; 88% agreement).**

1.17. In patients with cACLD and NITs in the indeterminate range (*i.e.* not meeting rule out or diagnostic criteria for CSPH), re-evaluation at 12 months may be the preferred management strategy **(LoE 4; weak recommendation; new; 88% agreement).**

1.18. Given the need to identify patients at substantial risk of decompensation, who benefit the most from preventive treatments (*e.g.* carvedilol/non-selective beta-blockers [NSBBs] and investigative therapies), the paradigm is shifting from non-invasive estimation of CSPH probability to non-invasive prediction of decompensation as a direct endpoint. Key prognostic indicators for decompensation include presence of CSPH, hepatic function, and aetiological factors **(LoE 2; new; 94% agreement).**

1.19. Patients with cACLD who are ineligible for NSBBs (contraindications or intolerance) to prevent decompensation should undergo oesophago-gastro-duodenoscopy (EGD) to screen for varices, unless LSM is <20 kPa and PLT ≥150 $\times 10^9$/L, or SSM <40 kPa **(LoE 1; strong recommendation; modified; 100% agreement).**

1.20. Patients with cACLD who are ineligible for NSBB therapy to prevent decompensation and who do not have varices at index EGD should undergo repeat EGD, unless they subsequently meet criteria specified in 1.19. Time intervals for repeating EGD are 2 years for cACLD patients with an active aetiological factor and 3 years for those in whom the primary aetiological factor has been removed/suppressed **(LoE 4; strong recommendation; new; 100% agreement).**

Use of NITs in patients with hepatocellular carcinoma

1.21. In patients with macrovascular invasion, NITs for CSPH/high-risk varices may not be applicable **(LoE 3; weak recommendation; new; 94% agreement).**

1.22. In patients with cACLD and Barcelona Clinic Liver Cancer (BCLC) stages 0-C, LSM ≤15 kPa and PLT ≥150$\times 10^9$/L rule out CSPH (negative predictive value >90%) and EGD may be avoided **(LoE 3; weak recommendation; new; 94% agreement).**

1.23. In patients with cACLD and BCLC stages 0-A without large (>5 cm) nodules in the right liver lobe, an estimated probability of CSPH ≥75% by ANTICIPATE/ANTICIPATE-NASH may be highly suggestive of CSPH **(LoE 3; weak recommendation; new; 95% agreement).**

Panel 1 - Paediatric statements

1.24. In children, extrahepatic portal vein obstruction and ACLD due to biliary atresia are the two leading causes of portal hypertension **(LoE 1; new; 95% agreement).**

1.25. LSM is a valuable complementary tool that enables the non-invasive identification of children at risk for portal hypertension due to ACLD and helps to monitor disease progression **(LoE 3; strong recommendation; new; 97% agreement).**

1.26. In paediatric patients, management of portal hypertension cannot rely solely on LSM and/or SSM, as cut-offs to identify the presence or absence of varices require further validation **(LoE 3; strong recommendation; new; 100% agreement).**

1.27. In children, integrating multiple NITs (*e.g.* the clinical prediction rule based on the age-adjusted Z-score for spleen length, PLT, and serum albumin) may facilitate prediction of high-risk varices and guide endoscopy intervals **(LoE 3; new; 100% agreement).**

Panel 1 – Research agenda

RA1.1. Validate non-VCTE elastography methods and VITRO for diagnosing CSPH and for prognostication.

RA1.2. Determine the utility of NITs to diagnose CSPH and prognosticate in rare liver disease aetiologies.

RA1.3. Confirm the utility of sequential NIT reassessments to guide patient management.

RA1.4. Investigate the use of NITs to monitor changes in HVPG.

RA1.5. Develop a generally accepted prognostic model for cACLD.

RA1.6. Integrate AI-based data for diagnosing CSPH and for prognostication.

RA1.7. Evaluate the added value of minimally invasive tests in patients with NITs in the indeterminate range (*i.e.* not meeting rule out or diagnostic criteria for CSPH).

RA1.8. Investigate the use of NITs to predict surgical risk (hepatic and extra-hepatic).

RA1.9. Define NIT-based criteria for (ruling out) varices in paediatric patients.

RA1.10. Develop NIT-based models for predicting bleeding in paediatric patients.

RA1.11. Investigate AI-based data integration for diagnosing varices and for prognostication in paediatric patients.

Panel 2 – Modifiers of steatotic liver disease

Alcohol abstinence and liver-related events

2.1. Return to/continuation of alcohol use is a major driver of liver-related events in patients with alcohol-related cACLD **(LoE 2; new; 98% agreement).**

2.2. In patients with alcohol-related cACLD, complete alcohol abstinence should be favoured over reduction of alcohol consumption to decrease liver-related events **(LoE 2; strong recommendation; new; 100% agreement).**

Interpretation of NIT changes in ALD

2.3. In patients with alcohol-related cACLD, a confirmed ≥30% relative change in LSM is associated with a corresponding change in the risk of liver-related events **(LoE 5; new; 94% agreement).**

2.4. In patients with alcohol-related liver disease (ALD), FIB-4, LSM, enhanced liver fibrosis (ELF) test and Fibrotest are associated with liver-related events and survival **(LoE 2; new; 100% agreement).**

Interpretation of NIT changes in MASLD

2.5. In MASLD-related cACLD, HVPG may underestimate the true portal pressure gradient. However, an HVPG ≥10 mmHg remains associated with a significantly increased risk of decompensation compared with the lower, albeit still

present, risk observed with HVPG <10 mmHg **(LoE 3; new; 98% agreement).**

2.6. In patients with MASLD-related cACLD without CSPH, a confirmed ≥30% relative reduction in LSM may indicate fibrosis regression of ≥1 stage **(LoE 3; new; 90% agreement).**

2.7. In patients with MASLD-related cACLD without CSPH, combining a confirmed ≥30% relative reduction in LSM with improvement in an independent non-invasive test (*e.g.* ≥0.5-unit absolute reduction in ELF) improves the accuracy of assessing regression of ≥1 fibrosis stage compared with LSM alone **(LoE 3; new; 88% agreement).**

2.8. NITs and their longitudinal changes should preferentially be calibrated to estimate the risk of liver-related events and changes in risk over time, rather than to assess fibrosis stage or changes in fibrosis stage **(LoE 4; new; 91% agreement).**

2.9. In patients with MASLD-related cACLD, a confirmed ≥30% relative change in LSM is associated with a clinically meaningful change in the risk of liver-related events, particularly when it results in crossing the 15 kPa risk threshold **(LoE 2; new; 96% agreement).**

2.10. Although the confirmed relative changes in LSM by point-shear wave elastography and 2D shear wave elastography likely carry similar clinical meaning as LSM, the available evidence is currently insufficient to support a specific recommendation on their use as monitoring tools **(LoE 4; new; 97% agreement).**

2.11. A ≥20% relative change in LSM by magnetic resonance elastography is associated with a clinically meaningful change in the risk of liver-related events in patients with MASLD-related cACLD **(LoE 4; new; 94% agreement).**

2.12. A 0.5-unit absolute change in ELF is associated with a clinically meaningful change in the risk of liver-related events in patients with MASLD **(LoE 3; new; 96% agreement).**

2.13. In patients with MASLD-related cACLD, a ≥10% weight reduction is associated with a ≥30% relative reduction in LSM, a surrogate of reduced risk of liver-related events **(LoE 3; new; 83% agreement).**

Metabolic and bariatric surgery

2.14. In patients with MASLD-related cACLD without CSPH, metabolic and bariatric surgery is associated with only a modest increase in the risk of surgical complications and may improve liver histology and liver-related events. Therefore, MASLD-related cACLD may be considered an obesity-associated morbidity that justifies consideration of metabolic and bariatric surgery at a lower BMI threshold **(LoE 3; new; 96% agreement).**

2.15. Surgical complications from metabolic and bariatric surgery are increased in patients with CSPH. Patients with features suggestive of CSPH who are being considered for metabolic and bariatric surgery should be managed by a multidisciplinary team at expert centres **(LoE 3; strong recommendation; new; 100% agreement).**

2.16. Metabolic and bariatric surgery is generally contraindicated in patients with decompensated cirrhosis, except in highly specialised settings, such as in conjunction with liver transplantation. These patients should be managed by a multidisciplinary team at expert centres **(LoE 3; strong recommendation; new; 98% agreement).**

Therapeutic agents for steatotic liver disease

2.17. There are currently no regulatory approved pharmacologic therapies for the treatment of patients with MASLD and established cirrhosis. Efficacy data derived from non-cirrhotic MASLD populations (moderate or advanced fibrosis) should not be extrapolated to patients with established cirrhosis **(LoE 2; strong recommendation; new; 98% agreement).**

2.18. In the absence of approved disease-modifying therapies, management of MASLD-related cirrhosis should focus on optimisation of lifestyle and metabolic comorbidities, prevention of hepatic decompensation, and referral for clinical trial participation **(LoE 3; strong recommendation; new; 100% agreement).**

2.19. SGLT2 inhibitors and GLP-1–based therapies, including mono- and multi-receptor agonists, can be safely used in patients with cACLD, in the absence of contraindications, for their approved indications, including type 2 diabetes and, for GLP-1–based therapies, obesity **(LoE 3; strong recommendation; new; 96% agreement).**

Panel 2 – Research agenda

RA2.1 Validate whether the most recent LSM result predicts liver-related events as accurately as longitudinal changes in LSM.

RA2.2 Determine the optimal strategy for repeating LSM to improve its reliability, including the timing of repeated examinations (*e.g.* at a different time on the same day or on a different day) and whether these examinations should be performed by the same or a different operator.

RA2.3 Assess whether calculating longitudinal changes in LSM using confirmed LSM, instead of single-point measurements, reduces intra-individual variability and allows adoption of a lower threshold for percentage change to define a true LSM response in SLD.

RA2.4 Determine the impact of obesity on LSM in SLD as well as across other aetiologies of chronic liver disease.

RA2.5 Establish the diagnostic and/or prognostic value of non-VCTE-based modalities to obtain LSM.

RA2.6 Determine how, and with what accuracy, longitudinal changes in LSM by different modalities correlate with changes in prognosis in patients with ALD.

RA2.7 Investigate the impact of weight loss, and the correlation between the magnitude of weight loss and its effect on portal hypertension (assessed invasively or non-invasively), separately in MASLD, MetALD, and in ALD with overweight/obesity.

RA2.8 Refine NITs and their thresholds for diagnosing CSPH specifically in the context of SLD, to more precisely predict the actual portal pressure gradient and/or the prognosis/risk of decompensation in patients with MASLD and obesity, and assess their sensitivity to change.

RA2.9 Evaluate the performance of NIT-based criteria for selecting appropriate patients in MASLD-cACLD/cirrhosis trials, with the aim of enriching the population for the desired endpoint(s) to improve trial feasibility.

RA2.10 Define a clinically meaningful change in NIT-based risk estimates in patients with SLD-related cACLD, to qualify these NITs as surrogate endpoints in clinical trials of SLD-related cACLD/cirrhosis.

Panel 3 – Prevention of decompensation

Definition of compensated cirrhosis/cACLD

3.1. Compensated cirrhosis/cACLD is defined by the absence of current or past decompensation of cirrhosis. The transition from compensated to decompensated cirrhosis markedly increases mortality risk (Fig. 2) **(LoE 2; modified; 99% agreement).**

3.2. a) Compensated cirrhosis/cACLD can be divided into two stages, based on the absence or presence of CSPH **(LoE 2).**

b) CSPH increases the risk of decompensation, especially in patients with gastro-oesophageal varices **(LoE 2).**

c) Patients with compensated cirrhosis/cACLD should be evaluated for the presence of CSPH **(LoE 2).**

d) The presence of varices and/or portosystemic collaterals are generally signs of CSPH **(LoE 2)**, but in cases of cured/controlled aetiology they may persist despite resolution of CSPH **(LoE 3).**

e) The main therapeutic goal in compensated cirrhosis/cACLD with CSPH is to prevent decompensation **(LoE 2; modified; 97% agreement).**

3.3. The events that define first decompensation are clinically evident ascites (or hepatic hydrothorax) caused by portal hypertension, variceal bleeding and/or overt hepatic encephalopathy (OHE) (West Haven grade ≥II) **(LoE 2; modified; 99% agreement).**

3.4. a) Minimal ascites or ascites detectable only by imaging **(LoE 3)**, covert HE **(LoE 3)** and occult bleeding from portal hypertensive gastroenteropathy **(LoE 5)** cannot be considered as decompensation.

b) Minimal ascites or ascites detectable only by imaging may be associated with decreased survival **(LoE 3; modified; 93% agreement).**

3.5. Jaundice alone (in non-cholestatic aetiologies) may be the first clinical manifestation in a minority of patients with cACLD/cirrhosis **(LoE 2)**; however, its definition, association with superimposed liver injury, and prognostic implications require further research **(LoE 4; modified; 96% agreement).**

Determinants of the natural history of compensated cirrhosis/cACLD

3.6. Development of hepatocellular carcinoma (HCC) in patients with cirrhosis/cACLD increases the risk of decompensation and/or death **(LoE 3; new; 99% agreement).**

3.7. Superimposed acute liver injury due to alcohol-related hepatitis, acute viral hepatitis (HEV, HAV, HBV), HBV flares, autoimmune hepatitis or drug-induced liver injury can precipitate decompensation and/or acute-on-chronic liver failure (ACLF) in patients with compensated cirrhosis/cACLD **(LoE 2; modified; 100% agreement).**

3.8. Diabetes mellitus and overweight/obesity are frequent comorbidities in patients with compensated cirrhosis/cACLD that can adversely impact prognosis and should be addressed in clinical trials and managed in clinical practice **(LoE 2; modified; 99% agreement).**

3.9. There is inconclusive data on the impact of sarcopenia and frailty on the natural history of compensated cirrhosis/cACLD **(LoE 2; modified; 94% agreement).**

3.10. Bacterial infections in compensated cirrhosis/cACLD patients with CSPH can precipitate decompensation and/or ACLF and, consequently, adversely affect natural history **(LoE 2; modified; 100% agreement).**

3.11. The prognostic impact of infections on the natural history of compensated cirrhosis/cACLD without CSPH is unclear **(LoE 5; modified; 96% agreement).**

3.12. Periodontal disease is associated with a higher risk of decompensation in compensated cirrhosis/cACLD **(LoE 4; new; 87% agreement).**

3.13. Surgery, especially major surgery, can precipitate decompensation and/or ACLF in patients with compensated cirrhosis/cACLD and CSPH, with the risk increasing as the severity of portal hypertension increases **(LoE 3; new; 100% agreement).**

Management and therapy

3.14. a) Successful aetiological treatment can improve portal hypertension and reduce the incidence of decompensation **(LoE 2).**

b) After aetiological therapy, CSPH can persist and should be reassessed **(LoE 3; strong recommendation; new; 99% agreement).**

3.15. Treatment with NSBBs, particularly carvedilol, is recommended in patients with compensated cirrhosis/cACLD and CSPH to prevent decompensation and improve survival **(LoE 2; strong recommendation; modified; 97% agreement).**

3.16. In patients with compensated cirrhosis/cACLD without CSPH, NSBBs should not be used for the prevention of decompensation/death **(LoE 2; strong recommendation; modified; 99% agreement).**

3.17. In patients with compensated cirrhosis/cACLD with CSPH and HCC, prevention of decompensation is a clinically relevant endpoint across all stages of the oncological disease, independent of tumour-directed therapy **(LoE 3; new; 100% agreement).**

3.18. Patients with compensated cirrhosis/cACLD who are on NSBBs do not require screening endoscopy for variceal detection as the results would not impact clinical management **(LoE 2; modified; 92% agreement).**

3.19. In patients with compensated cirrhosis/cACLD with contraindications or intolerance to NSBBs, endoscopy is recommended if CSPH cannot be ruled-out and if large varices (oesophageal varices [OV] and type 1 gastro-oesophageal varices [GOV1]) are detected, endoscopic variceal ligation (EVL) is recommended to prevent first variceal bleeding **(LoE 1; strong recommendation).** Endoscopic therapy on its own has no effect on prevention of non-bleeding decompensating events **(LoE 1; modified; 100% agreement).**

3.20 a) In compensated cirrhosis/cACLD patients with CSPH, transjugular intrahepatic portosystemic shunt (TIPS) placement is not recommended for the prevention of decompensation **(LoE 5; strong recommendation).**

b) Preoperative TIPS placement may be considered prior to major surgery to reduce the risk of postoperative decompensation **(LoE 4; weak recommendation; modified; 97% agreement).**

3.21.a) Patients with compensated cirrhosis/cACLD with gastric varices have CSPH and benefit from NSBBs, particularly carvedilol, for the prevention of decompensation **(LoE 2; modified; 96% agreement).**

b) In patients with high risk GOV2 or IGV1 (cardiofundal) varices who have contraindications or intolerance to NSBBs, local therapy (*i.e.* endoscopic injection therapy **(LoE 2; strong recommendation)**, endoscopic ultrasound [EUS]-guided therapy **(LoE 4; weak recommendation)** or transvenous obliteration **(LoE 3; weak recommendation)**) may be considered to prevent bleeding in centres with expertise **(modified; 95% agreement).**

c) In patients with compensated cirrhosis/cACLD and GOV2 or IGV1 (cardiofundal) varices, there is limited evidence to support the use of endoscopic injection therapy **(LoE 3)**, EUS-guided therapy **(LoE 5)** or transvenous obliteration **(LoE 5)** in addition to NSBBs to prevent first variceal bleeding and improve survival **(modified; 96% agreement).**

3.22. There is insufficient evidence to recommend statins or anticoagulation to prevent decompensation **(LoE 3)** but they should be used or maintained if prescribed for approved indications **(LoE 2; strong recommendation; modified; 99% agreement).**

3.23. Regular dental and periodontal care may decrease the risk of decompensation in compensated cirrhosis/cACLD **(LoE 3; weak recommendation; new; 81% agreement).**

Panel 3 – Research agenda

Definition

RA3.1. Define and determine the prognostic impact of jaundice alone as a first decompensating event.

RA3.2. Define HE and its differential diagnosis as a first decompensating event.

RA3.3. Determine the prognostic significance of minimal ascites (only detected on imaging), minimal hepatic encephalopathy, or chronic bleeding from portal hypertensive gastropathy (PHG) in patients with cACLD.

RA3.4. Evaluate the impact of different types of first decompensating events and their acuity on mortality.

Determinants of natural history

RA3.5. Further characterise the interaction between HCC and portal hypertension on the natural history of cACLD.

RA3.6. Evaluate the impact of comorbidities, including their early detection and treatment, on the risk of liver-related events and on competing non–liver-related mortality (*e.g.* cardiovascular disease, chronic kidney disease, chronic obstructive pulmonary disease, non-hepatic malignancies) in patients with cACLD.

RA3.7. Evaluate the impact of multimorbidity on the management of patients with compensated cirrhosis.

RA3.8. Evaluate PVT as a precipitant of first decompensation.

RA3.9. Determine the impact of sarcopenia and frailty (and their treatment) on the natural history of cACLD.

RA3.10. Evaluate the incidence and prognosis of (bacterial) infections in patients with cACLD with and without CSPH.

RA3.11. Evaluate whether prevention of infections (*i.e.* vaccination, oral/dental health) has an impact on the incidence of decompensation.

RA3.12. Evaluate the impact of health measures and dedicated care bundles (including management of sarcopenia, nutritional counselling, physical activity, extrahepatic cancer screening, bone health assessment, vaccination, oral/dental health assessment and counselling on concomitant medication including potential hepatotoxic drugs) on the natural history of compensated cirrhosis/cACLD.

Management and therapy

RA3.13. Evaluate the prevention of decompensation with NSBBs in patients with CSPH without varices.

RA3.14. Evaluate the role of statins in preventing decompensation.

RA3.15. Evaluate the use of local therapies (endoscopic, endosonographic, or transvenous obliteration), in addition to NSBBs, in preventing first gastric variceal bleeding.

RA3.16. Evaluate the role of pre-surgical TIPS in preventing post-surgical decompensation in patients with cACLD.

RA3.17. Evaluate other therapeutic targets (incretin-based drugs, FGF21 analogues, anticoagulation/antiplatelet agents, antifibrotics like galectin-3 inhibitors [*e.g.* belapectin] or prolonged-release pirfenidone) for preventing decompensation.

Panel 4 – Prevention of further decompensation

4.1. In patients with decompensated cirrhosis, further decompensation (FD) represents a more advanced disease stage with a higher mortality **(LoE 2; modified; 86% agreement)** and is defined as follows (Figure 2):

(a) In patients with a single decompensation:

- *If ascites*, by the need for ≥3 large-volume paracenteses within 1 year, the occurrence of spontaneous bacterial peritonitis (SBP), hepatorenal syndrome-acute kidney injury (HRS-AKI), or a second decompensating event (CSPH-related bleeding, OHE, or jaundice);
- *If variceal bleeding*, by the development of CSPH-related re-bleeding or a second decompensating event (ascites, OHE, or jaundice);
- *If OHE,* by the development of recurrent OHE or a second decompensating event (ascites, bleeding, or jaundice).

(b) In patients with ≥2 decompensating events, by any combination of ascites, portal hypertension-related bleeding, OHE and/or jaundice.

4.2. After the first decompensating event, an aetiological treatment is recommended to prevent FD **(LoE 2; strong recommendation; new; 100% agreement).**

4.3. Patients with decompensated cirrhosis should be considered for liver transplantation (LT) **(LoE 1; strong recommendation; unchanged; 98% agreement).**

4.4. In patients with ascites and varices, carvedilol/cNSBBs should be used to prevent first bleeding **(LoE 1; strong recommendation)**, and to lower the risk of ascites-related

complications **(LoE 3; strong recommendation),** with carvedilol being preferred over cNSBBs **(LoE 3; strong recommendation; new; 81% agreement).**

4.5. In patients with ascites and varices, ineligible for carvedilol/cNSBBs (intolerance or contraindication), endoscopic therapy alone should be considered to prevent first variceal bleeding **(LoE 3; strong recommendation; modified; 92% agreement).**

4.6. Carvedilol/cNSBBs should be dose-reduced or discontinued in case of persistently low blood pressure (systolic arterial pressure <90 mmHg or mean arterial pressure <65 mmHg) and/or HRS-AKI **(LoE 2; strong recommendation; modified; 95% agreement).**

4.7. Carvedilol/cNSBBs should be re-initiated and/or re-titrated once HRS-AKI has resolved and/or blood pressure has recovered **(LoE 2; strong recommendation; 95% agreement).**

4.8. In patients with a history of bleeding from OV as the first decompensating event, the first-line therapy for prevention of recurrent bleeding is the combination of NSBB+EVL **(LoE 1; strong recommendation)**, with carvedilol being preferred over cNSBBs **(LoE 2; strong recommendation)**, independently of prior primary prophylaxis with carvedilol/cNSBBs **(modified; 85% agreement).**

4.9. In patients recovering from an episode of acute OV bleeding as their single decompensating event (ineligible for rescue or pre-emptive TIPS [pTIPS]) who cannot receive either EVL or carvedilol/cNSBBs, the available tolerated therapy should be initiated or maintained as monotherapy **(LoE 1; strong recommendation; modified; 87% agreement).**

4.10. In patients recovering from an episode of acute bleeding from OV (ineligible for rescue or pTIPS) who also experienced a non-bleeding decompensating event, TIPS and LT may be considered **(LoE 3; weak recommendation; modified; 92% agreement).**

4.11. In patients with decompensated cirrhosis, statins are not recommended to prevent FD, but they should be used or continued if prescribed for approved indications, considering the higher risk of statin toxicity in patients with Child-Pugh C cirrhosis **(LoE 2; strong recommendation; modified; 82% agreement).**

4.12. In patients after the first decompensating event, long-term primary antibiotic prophylaxis is not recommended to prevent FD **(LoE 2; strong recommendation; new; 99% agreement).**

4.13. TIPS should be considered in patients with portal hypertension-related bleeding despite secondary prophylaxis with carvedilol/cNSBB+EVL (failure of secondary prophylaxis) **(LoE 3; strong recommendation; modified; 92% agreement).**

4.14. The prevention of FD in patients with HCC should be managed in the same way as in patients without HCC, while considering life expectancy and treatment side effects **(LoE 3; strong recommendation; new; 95% agreement).**

4.15. The presence of HCC does not represent an absolute contraindication to TIPS in patients with recurrent/refractory ascites or hepatic hydrothorax, unless the tumour lies along the TIPS trajectory. An improved control of hepatic decompensation with TIPS may facilitate access to effective oncological treatment **(LoE 3; weak recommendation; new; 90% agreement).**

4.16. Frailty, malnutrition, sarcopenia and myosteatosis have an impact on survival in patients with decompensated cirrhosis or a prior history of decompensation **(LoE 2; modified).** These conditions should be evaluated using available standardised tools (see Table S1) **(LoE 3; strong recommendation; unchanged; 90% agreement).**

4.17. All patients with decompensated cirrhosis should undergo a structured nutritional assessment, including an individualised nutrition plan and counselling on regular physical activity **(LoE 2; strong recommendation; modified; 94% agreement).**

4.18. In patients identified as frail and/or sarcopenic, targeted interventions should be encouraged in addition to standard nutritional and physical activity counselling, as muscle health may improve and lead to improved clinical outcomes **(LoE 2; strong recommendation),** but further evaluation is required regarding their impact on the natural history of further decompensation **(LoE 3; weak recommendation; new; 92% agreement).**

4.19. Sarcopenia and frailty can improve after TIPS but can also be associated with poor outcomes. Sarcopenia and/or frailty alone should neither indicate nor contraindicate TIPS but should be considered together with other factors in the pre-TIPS work-up **(LoE 3; strong recommendation; modified; 91% agreement).**

Panel 4 – Research agenda

RA4.1. Validate the prognostic weight of any decompensating event occurring within 5 days after acute variceal bleeding.

RA4.2. Assess the prognostic impact of Grade 1 (mild) ascites as the first decompensating event.

RA4.3. Investigate the effect of time to FD on prognosis in patients with ascites/OHE as first decompensation.

RA4.4. Determine the role of inflammatory biomarkers on predicting the risk of FD.

RA4.5. Integrate the prognostic concept of FD into clinical decision algorithms to optimise patient selection for more advanced treatments (*e.g.* TIPS or other interventional radiology, LT).

RA4.6. Analyse the positioning of the definition of FD and possible overlap with the concepts and definitions of non-acute, acute decompensation and ACLF.

RA4.7. Investigate whether prevention of FD in patients with ascites should be tailored according to the aetiology of cirrhosis and comorbidities (*i.e.* patients with MASLD and/or diabetes).

RA4.8. Determine how the type and severity of hepatic decompensation at the time of aetiological therapy influences subsequent risk of progression to FD.

RA4.9. Identify *a priori* patients in whom FD cannot be prevented despite effective aetiological therapy.

RA4.10. Assess whether carvedilol/cNSSBs should be started in patients with ascites as first decompensation regardless of the presence of varices.

RA4.11. Investigate the effectiveness of EVL in addition to carvedilol/cNSBBs for primary prophylaxis of variceal bleeding in patients with a history of first non-bleeding event.

RA4.12. Identify the subgroup of patients with ascites who benefit from long-term albumin administration, investigate the optimal baseline serum albumin level for treatment initiation, and determine criteria for continued benefit *vs.* futility of this treatment.

RA4.13. Perform randomised controlled trials (RCTs) comparing efficacy, tolerance and costs of long-term albumin administration *vs.* TIPS in patients who are candidates for both treatments.
RA4.14. Assess if statins, metformin, or GLP-1 receptor agonists prevent FD in certain populations of patients with decompensated cirrhosis in prospective studies.
RA4.15. Obtain confirmatory data on primary prophylaxis of SBP in high-risk patients, carefully balancing efficacy against the risk of multidrug-resistant organism infections.
RA4.16. Evaluate whether elective TIPS placement can improve outcomes in patients who do not meet criteria for pTIPS at the time of bleeding but who are nonetheless identified as being at increased risk of FD and death.
RA4.17. Explore whether elective TIPS placement can improve outcomes in patients who are eligible for pTIPS but did not receive pTIPS during the 72 h time window or in patients with a first episode of tense ascites.
RA4.18. Investigate the effects of TIPS on quality of life and on access to HCC treatment according to BCLC stage.
RA4.19. Assess the additive value of body composition parameters in addition to Child-Pugh/MELD scores.
RA4.20. Evaluate the impact of sarcopenia and/or frailty and its management on FD.
RA4.21. Investigate, in prospective studies, the impact of sarcopenia and/or frailty on post-TIPS outcomes.
RA4.22. Explore the impact of anticoagulation (type, dose and target population) on preventing FD.
RA4.23. Investigate the effect of carvedilol/cNSBBs on preventing FD in patients with ascites and PHG.

Panel 5 – Management of portal hypertensive bleeding

5.1. Acute variceal bleeding (AVB) is defined by bleeding from oesophageal, gastric or ectopic varices (blood emanating from a varix: spurting or oozing), or by varices with signs of recent bleeding (clots, white nipple sign), or by varices without other sources of bleeding in the presence of blood in the stomach in a patient with upper gastrointestinal bleeding **(LoE 4; modified; 85% agreement).**
5.2. Active bleeding at endoscopy is defined as blood coming from a varix (spurting, oozing). However, interobserver agreement for the diagnosis of active bleeding is moderate **(LoE 3; modified; 87% agreement).**
5.3. The goal of resuscitation is to preserve organ perfusion. Volume restitution should aim to restore and maintain haemodynamic stability (Fig. 3) **(LoE 5; strong recommendation; modified; 97% agreement).**
5.4. Restrictive transfusion of packed red blood cell transfusions should be performed with a target haemoglobin level between 7-8 g/dl. However, individualised transfusion strategies should be considered based on the presence of cardiovascular disorders, age, haemodynamic status, and ongoing bleeding **(LoE 2; strong recommendation; modified; 92% agreement).**
5.5. Intubation is recommended before endoscopy in patients with altered consciousness or active haematemesis **(LoE 5; strong recommendation; modified; 97% agreement).**
5.6. Extubation should be performed as quickly as possible after endoscopy **(LoE 5; strong recommendation; unchanged; 96% agreement).**
5.7. In suspected variceal bleeding, vasoactive drugs (terlipressin, somatostatin, octreotide) should be started as soon as possible and continued for 2-5 days or until a TIPS (if indicated) is placed **(LoE 1; strong recommendation; modified; 82% agreement).**
5.8. A shorter course (even 24 h) of vasoactive drugs can be considered following successful endoscopic haemostasis, provided NSBB treatment is then initiated **(LoE 2; strong recommendation; modified; 82% agreement).**
5.9. Sodium levels should be monitored for patients on terlipressin due to risk of hyponatraemia **(LoE 2; strong recommendation; modified; 97% agreement).**
5.10. Preventive antibiotic therapy is recommended at presentation with AVB **(LoE 1; strong recommendation).** Duration and need may be individualised based on infection risk and cirrhosis severity, as patients with Child–Pugh A cirrhosis may only require a shorter course or no therapy **(LoE 2; weak recommendation; modified; 85% agreement).**
5.11. In patients with cirrhosis experiencing AVB, third generation cephalosporins are generally preferred for preventive antibiotic therapy; however, local resistance patterns and antimicrobial policies should be considered **(LoE 2; strong recommendation; modified; 94% agreement).**
5.12. Malnutrition increases the risk of adverse outcomes in patients with cirrhosis and AVB. Therefore, oral nutrition should be started as soon as possible **(LoE 5; strong recommendation; modified; 93% agreement).**
5.13. Airway manipulation and placement of a nasogastric tube should be performed with caution because of the risk of aspiration and pulmonary infection **(LoE 5; strong recommendation; modified; 91% agreement).**
5.14. Proton pump inhibitors (PPIs), when started before endoscopy, should be discontinued after portal hypertension-related bleeding has been confirmed, unless there is a strong indication to continue them **(LoE 5; strong recommendation; modified; 91% agreement).**
5.15. Six-week mortality is the recommended primary endpoint for studies on the treatment of AVB **(LoE 5; strong recommendation; modified; 97% agreement).**
5.16. Treatment failure is defined either by uncontrolled bleeding or by rebleeding within 5 days **(LoE 5; modified; 94% agreement).**
5.17. Child-Pugh class C, MELD and its variants (*e.g.* MELD-Na, MELD 3.0), and failure to control bleeding represent key risk factors for 6-week mortality **(LoE 1; modified; 87% agreement).**
5.18. Following haemodynamic resuscitation, patients with suspected AVB should undergo upper endoscopy within 12 h of presentation **(LoE 2; strong recommendation).** If the patient is haemodynamically unstable, endoscopy should be performed as soon as possible depending on local availability **(LoE 5; strong recommendation; modified; 93% agreement).**
5.19. In centres with a hepatology service, 24/7 availability of an on-call gastrointestinal (GI) endoscopist and assistant staff proficient in endoscopic treatment of variceal/GI bleeding is

recommended **(LoE 5; strong recommendation; modified; 94% agreement).**

5.20. In patients with suspected AVB and no contraindications (QT prolongation), a single intravenous dose of erythromycin (250 mg, 30–90 min before endoscopy) may improve gastric clearance, enhance endoscopic visualisation and result in a reduced blood transfusion requirement **(LoE 2; weak recommendation; new; 94% agreement).**

5.21. Patients with AVB should be managed in intensive or intermediate care units **(LoE 5; strong recommendation; unchanged; 94% agreement).**

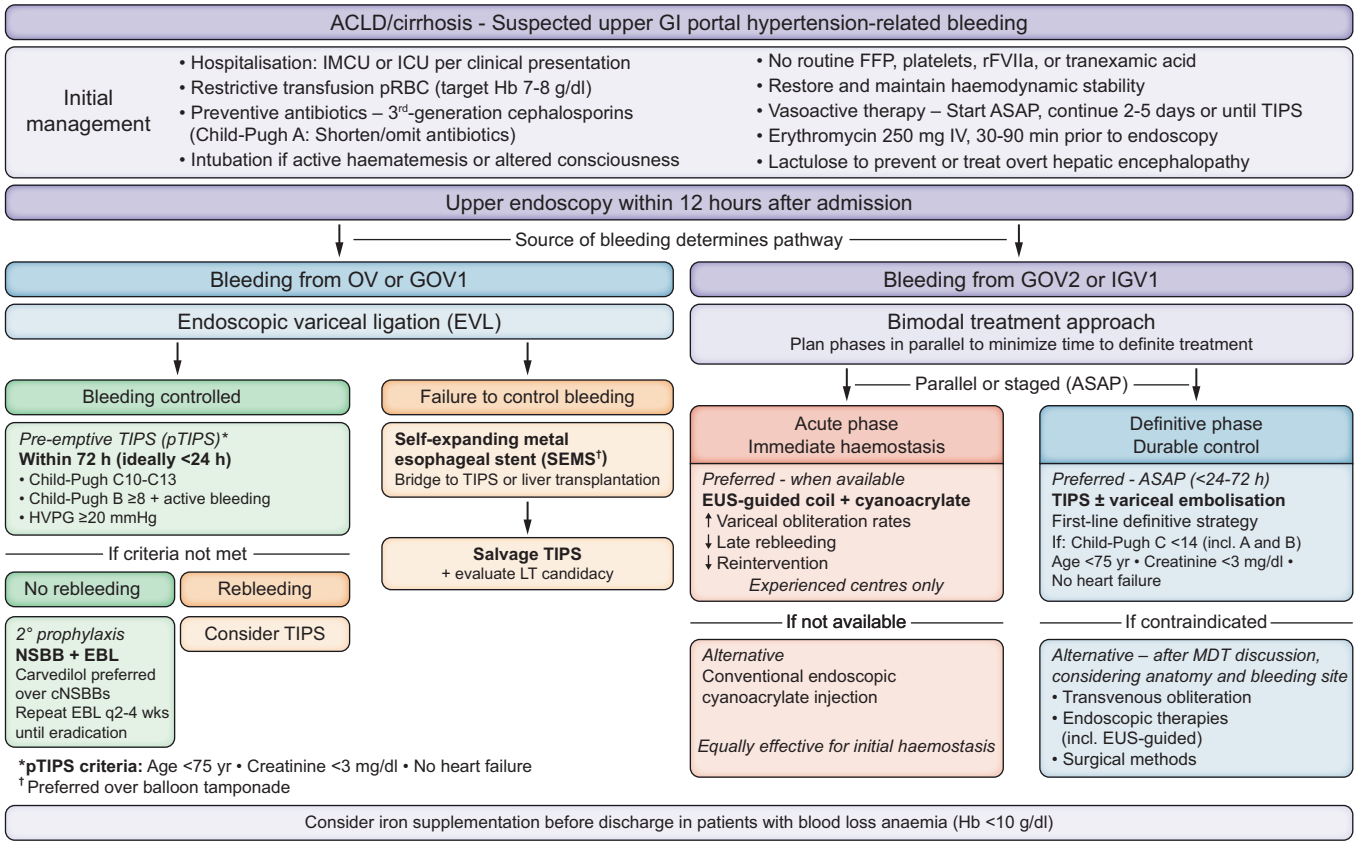


Fig. 3. Baveno-VIII algorithm for the management of acute PH bleeding. ACLD, advanced chronic liver disease; ASAP, as soon as possible; EBL, endoscopic band ligation; EUS, endoscopic ultrasound; FFP, fresh frozen plasma; GOV, gastro-oesophageal varices; HVPG, hepatic venous pressure gradient; ICU, intensive care unit; IMCU, intermediate care unit; IGV, isolated gastric varices; MDT, multidisciplinary team; NSBBs, non-selective beta blockers; OV, oesophageal varices; pRBC, packed red blood cells; (p)TIPS, (pre-emptive) transjugular portosystemic shunt

5.22. In patients presenting with AVB, lactulose is recommended (orally or as an enema) to prevent or treat OHE by accelerating the clearance of blood from the GI tract **(LoE 2; strong recommendation; modified; 91% agreement).**

5.23. In patients with cirrhosis and AVB, transfusion of fresh frozen plasma is not routinely recommended, as it does not correct coagulopathy and may lead to volume overload and worsening of portal hypertension **(LoE 4; strong recommendation; modified; 96% agreement).**

5.24. In patients with cirrhosis and AVB, platelet count and fibrinogen levels do not correlate with the risk of failure to control bleeding or rebleeding and their transfusion is not routinely recommended. However, in case of failure to control bleeding, transfusion of platelets or fibrinogen supplementation may be considered **(LoE 5; weak recommendation; modified; 90% agreement).**

5.25. The use of recombinant factor VIIa and tranexamic acid is not recommended for AVB **(LoE 2; strong recommendation; modified; 94% agreement).**

5.26. In patients with AVB, anticoagulation should be temporarily discontinued until bleeding is controlled. Timing of re-initiation should be individualised based on the strength of the indication for anticoagulation **(LoE 5; strong recommendation; modified; 96% agreement).**

5.27. In patients recovering from AVB, iron supplementation should be considered in those with anaemia due to blood loss **(LoE 3; strong recommendation; new; 87% agreement).**

5.28. The use of haemoclips, over-the-scope-clips, topical haemostatic agents and fibrin is not recommended as first-line endoscopic therapy for AVB **(LoE 4; strong recommendation; modified; 97% agreement).**

5.29. In AVB, immediate haemostasis with EVL is recommended for AVB from oesophageal varices and GOV1 **(LoE 1; strong recommendation; modified; 96% agreement).**

5.30. All patients with AVB should undergo abdominal imaging, preferably contrast-enhanced cross-sectional imaging (CT or MRI) to exclude portal/splanchnic vein thrombosis and HCC, and to map portosystemic collaterals in order to guide

treatment **(LoE 5; strong recommendation; modified; 87% agreement).**

5.31. pTIPS with polytetrafluoroethylene-covered stents within 72 h (ideally <24 h) is indicated in patients bleeding from OV and GOV1 who meet any of the following criteria: Child-Pugh class C 10-13 points or Child-Pugh class B >7 with active bleeding at initial endoscopy or HVPG ≥20 mmHg at the time of AVB **(LoE 1; strong recommendation; modified; 93% agreement).**

5.32. In patients fulfilling the pTIPS criteria who did not undergo TIPS within the preferred 72h window, TIPS placement within 2 weeks in patients with Child-Pugh class C 10-13 or within 1 week in patients with Child-Pugh class B >7 with active bleeding may still provide benefit **(LoE 4; weak recommendation; new; 88% agreement).**

5.33. In patients fulfilling pTIPS criteria, ACLF, OHE, hyperbilirubinaemia, MELD score, or severe alcohol-related hepatitis should not be considered absolute contraindications **(LoE 3; strong recommendation; new; 94% agreement).**

5.34. Patients with OHE refractory to medical and interventional therapy after TIPS may be prioritised for LT **(LoE 4; weak recommendation; new; 85% agreement).**

5.35. In refractory oesophageal variceal bleeding, a dedicated covered self-expanding metal stent is preferred over balloon tamponade, as bridging therapy to TIPS or LT **(LoE 4; strong recommendation; modified; 91% agreement).**

5.36. Failure to control variceal bleeding despite combined pharmacological and endoscopic therapy should be managed by salvage TIPS **(LoE 2; strong recommendation; modified; 97% agreement).**

5.37. Salvage TIPS should be considered in all patients with refractory variceal bleeding, regardless of age, Child-Pugh class, or MELD score, on a case-by-case basis **(LoE 3; strong recommendation; modified; 96% agreement).**

5.38. The Sarin classification is recommended for gastric varices, distinguishing GOV1/2 and IGV1/2 in order to harmonise reporting, stratify bleeding risk, and inform therapeutic strategy **(LoE 3; strong recommendation; new; 97% agreement).**

5.39. AVB from GOV2 or IGV1 is a high-risk event that requires rapid, structured multidisciplinary management. Treatment is bimodal: an acute phase focused on immediate haemostasis and a definitive phase, pursuing durable bleeding control and prevention of FD. Both phases may start in parallel; if staged – the interval between them should be kept as short as possible **(LoE 5; strong recommendation; new; 87% agreement).**

5.40. In patients with AVB from GOV2 or IGV1, immediate haemostasis can be achieved with either conventional endoscopic cyanoacrylate injection or EUS-guided therapy, as both appear similarly effective in controlling bleeding and preventing early rebleeding. When available, EUS-guided coil plus cyanoacrylate may be recommended in experienced centres because it may achieve higher variceal obliteration rates, and reduce late rebleeding and the need for reintervention **(LoE 2; weak recommendation; new; 81% agreement).**

5.41. TIPS ± variceal embolisation performed as soon as possible is the preferred definitive treatment option in patients with AVB from GOV2/IGV1. When TIPS is contraindicated, other definitive strategies – including transvenous obliteration, endoscopic therapies (including EUS-guided approaches), or surgical methods – may be utilised depending on available expertise and after multidisciplinary discussion based on the anatomy and location of the bleeding site **(LoE 2; strong recommendation; new; 81% agreement).**

5.42. Either endovascular (TIPS or transvenous obliteration) or endoscopic treatment should be considered in patients with ectopic varices **(LoE 4; strong recommendation; modified; 91% agreement).**

5.43. In patients with gastric antral vascular ectasia requiring serial transfusion, EVL is recommended over argon plasma coagulation, as it is associated with higher eradication rates and fewer treatment sessions, with reductions in recurrent bleeding, hospitalisations, and transfusion requirements **(LoE 1; strong recommendation; modified; 90% agreement).**

5.44. Management of PHG focuses primarily on reducing portal pressure with NSBBs. In cases of severe or refractory bleeding from PHG, TIPS is the most effective rescue therapy. Since endoscopic therapies do not improve portal hypertension, they provide limited benefit and are reserved for focal bleeding lesions or salvage situations **(LoE 4; modified; 91% agreement).**

5.45. In patients with cirrhosis and PVT, management of AVB should generally follow the recommendations for patients without PVT. In selected patients, TIPS ± portal vein recanalisation may be considered when additional factors related to portal hypertension are present (*e.g.* ascites or documented progression of PVT) or LT candidacy **(LoE 5; strong recommendation; modified; 85% agreement).**

5.46. The presence of HCC alone may not preclude consideration of all potential treatments in the setting of AVB. Treatment decisions should be individualised within a multidisciplinary framework and should account for futility criteria related to both liver function and tumour prognosis **(LoE 5; weak recommendation; new; 94% agreement).**

5.47. Post-EVL ulcer bleeding is a rare but serious complication, the management of which should be individualised and may require vasoactive therapy, endoscopic haemostasis, temporary stenting or tamponade, and rescue TIPS with embolisation when needed, particularly in high-risk patients (MELD >18 or emergency EVL) **(LoE 1; strong recommendation; new; 88% agreement).**

5.48. Current evidence is insufficient to recommend the routine use of PPIs for the prevention of (bleeding from) post-EVL ulcers **(LoE 2; new; 91% agreement).**

Panel 5 – Research agenda

RA5.1. Evaluate interobserver agreement for AVB.

RA5.2. Compare preventive antibiotic therapy *vs*. no antibiotics in Child-Pugh A patients with AVB in an RCT.

RA5.3. Validate a short course (under 48 h) of vasoactive therapy (including type of drug) in Child-Pugh A and B7 patients with AVB.

RA5.4. Compare continuous vasoactive infusion with bolus administration in terms of AVB control and associated adverse effects.

RA5.5. Assess the efficacy and safety of tranexamic acid in the management of AVB.

RA5.6. Validate high-risk criteria for pTIPS including CLIF-C AD and the Chinese Collaboration Group AVB.

RA5.7. Evaluate the impact of a 1- or 2-week delay to pTIPS *vs*. NSBB+EVL in patients with Child-Pugh C or Child-Pugh B ≥8 + active bleeding.
RA5.8. Evaluate the impact of PVT on outcomes of pre-emptive TIPS in patients with cirrhosis and AVB.
RA5.9. Assess whether pre-emptive TIPS confers a survival benefit in patients with AVB and HCC beyond Milan criteria.
RA5.10. Investigate the role of pre-emptive TIPS in patients with AVB aged >75 years, incorporating frailty assessment as a stratification variable.
RA5.11. Assess rifaximin as primary prophylaxis for OHE before pTIPS.
RA5.12. Investigate the effectiveness of underdilated/small pTIPS in AVB.
RA5.13. Build and externally validate a new futility score for rescue TIPS.
RA5.14. Investigate the effectiveness of TIPS for bleeding from PHG.
RA5.15. Perform an RCT comparing TIPS/transvenous obliteration and EUS-guided therapies for the definitive (long-term) management of gastric varices (GOV2/IGV1).
RA5.16. Determine the optimal strategies to prevent and manage bleeding from post-EVL ulcers.

Panel 6 – Management of further decompensation

6.1. Patients with further decompensation (FD) should be evaluated for LT **(LoE 1; strong recommendation; new; 97% agreement).**
6.2. In patients with FD, aetiological cure/control improves survival and should be pursued **(LoE 2; strong recommendation; new; 96% agreement).**
6.3. Patients with recurrent or refractory ascites should be considered for TIPS **(LoE 1; strong recommendation; modified; 97% agreement).**
6.4. In patients with recurrent or refractory ascites/hydrothorax, TIPS and LT should be discussed together **(LoE 2; strong recommendation; new; 91% agreement).**
6.5. In patients with recurrent/refractory ascites, to reduce the risk of post-TIPS overshunting (HE, cardiac failure) while preserving efficacy, the stent may be dilated to the smallest diameter that achieves an adequate response **(LoE 3; weak recommendation; new; 97% agreement).**
6.6. Stepwise dilation should be considered in patients with no improvement in ascites **(LoE 3; strong recommendation; new; 97% agreement).**
6.7. In patients with hepatic hydrothorax refractory to standard medical treatment, TIPS should be considered **(LoE 3; strong recommendation; new; 90% agreement).**
6.8. TIPS may be considered for the management of recurrent/refractory ascites or refractory hepatic hydrothorax in selected older adults (≥70 years) on a case-by-case basis and taking into account the benefit/risk ratio **(LoE 3; weak recommendation; new; 90% agreement).**
6.9. In patients undergoing evaluation for TIPS for recurrent/refractory ascites, renal function should be systematically assessed, and chronic kidney disease staged (estimated glomerular filtration rate) for risk stratification **(LoE 2; strong recommendation; new; 91% agreement).**
6.10. In patients with recurrent/refractory ascites who are not TIPS candidates, long-term albumin treatment may be considered as it can improve ascites control and reduce the incidence of ascites-related complications **(LoE 3; weak recommendation; new; 83% agreement).**
6.11. In patients with refractory ascites who have contraindications or an inadequate response to TIPS and are not candidates for LT, home-based drainage devices (tunnelled peritoneal catheter or low-flow ascites pump) may be considered as an alternative to repeated large-volume paracentesis **(LoE 3; weak recommendation; new; 93% agreement).**
6.12. In patients with recurrent/persistent HE despite optimal pharmacological therapy, embolisation of large portosystemic shunts should be considered, in addition to LT discussion, particularly when liver function is preserved **(LoE 3; strong recommendation; new; 91% agreement).**
6.13. In patients with recurrent/persistent HE, treatment with L-ornithine-L-aspartate (LOLA) can be considered **(LoE 3; weak recommendation; new; 80% agreement).**
6.14. Before elective TIPS placement, history of OHE and use of HE medications should be carefully assessed **(LoE 2; strong recommendation; new; 96% agreement).**
6.15. A history of prior OHE is not an absolute contraindication for elective TIPS placement, and the decision should be individualised based on the patient's clinical context **(LoE 2; strong recommendation; new; 95% agreement).**
6.16. Rifaximin should be considered for prophylaxis of HE in patients with cirrhosis who are candidates for elective TIPS placement **(LoE 2; strong recommendation; modified; 85% agreement).**
6.17. In patients with cirrhosis who develop recurrent or persistent HE after TIPS despite removal of precipitants and standard medical treatment, spontaneous shunt embolisation and/or reduction or occlusion of the TIPS may be considered **(LoE 3; weak recommendation; new; 87% agreement).**
6.18. In patients with HRS-AKI, treatment with terlipressin and albumin should be considered **(LoE 1; strong recommendation; new; 93% agreement).**
6.19. Terlipressin should preferably be administered as continuous intravenous infusion (starting dose 2-3 mg/24 h) to reduce the incidence of adverse events **(LoE 2; strong recommendation; new; 93% agreement).**
6.20. Norepinephrine can be used as an alternative to terlipressin, if terlipressin is unavailable, contraindicated or not tolerated **(LoE 1; strong recommendation; new; 86% agreement).**
6.21. In patients with cirrhosis and a history of SBP, antibiotic prophylaxis should be considered to prevent recurrence of SBP **(LoE 2; strong recommendation; modified; 93% agreement).**
6.22. Antibiotic SBP prophylaxis should be discontinued in case of recompensation **(LoE 4; strong recommendation; new; 99% agreement).**
6.23. Data are insufficient to recommend rifaximin for secondary prophylaxis of SBP **(LoE 4; new; 94% agreement).**
6.24. In patients with FD, withdrawal of PPIs is recommended unless there is a clear indication **(LoE 2; strong recommendation; new; 93% agreement).**
6.25. In patients with jaundice as the sole manifestation of FD, an immediate and exhaustive search for precipitating events, with particular attention to bacterial infections (even if

asymptomatic), alcohol-related hepatitis, flares of viral or autoimmune hepatitis, drug-induced liver injury, or biliary obstruction, should be performed **(LoE 3; strong recommendation; new; 86% agreement).**

Panel 6 – Research agenda

RA6.1. Define trajectories of FD according to the type and number of first FD event(s).
RA6.2. Define the clinical trajectories of FD according to whether it presents acutely or non-acutely.
RA6.3. Define trajectories of FD according to the cause of chronic liver disease and the possibility of eliminating/controlling the aetiological factor.
RA6.4. Identify the optimal haemodynamic response for patients with recurrent/refractory ascites undergoing TIPS placement.
RA6.5. Define how to assess haemodynamic response.
RA6.6. Develop a holistic approach to TIPS in patients with recurrent or refractory ascites, integrating splanchnic and systemic haemodynamic data, sarcopenia and frailty, systemic and local inflammatory status, gut microbiota, and comorbidities.
RA6.7. Assess, in an RCT, the efficacy of small diameter TIPS in recurrent/refractory ascites and hydrothorax.
RA6.8. Evaluate the role of small-diameter TIPS in older adult populations and identify predictors of outcome.
RA6.9. Evaluate the role of prehabilitation and post-TIPS rehabilitation programmes in patients undergoing elective TIPS.
RA6.10. Evaluate strategies to prevent congestive heart failure after TIPS.
RA6.11. Assess the role of faecal microbiota transplant (FMT) in the prevention of HE.
RA6.12. Assess the role of FMT in the secondary prevention of HE (phase III trial).
RA6.13. Assess the role of rifaximin prophylaxis at the time of preemptive/salvage/rescue TIPS placement in the prevention of HE.
RA6.14. Identify the optimal timing for discontinuation of rifaximin after TIPS.
RA6.15. Evaluate the potential role of rifaximin in promoting the emergence of multidrug-resistant organism colonisation.
RA6.16. Assess the use of FMT in patients with recurrent/refractory HE.
RA6.17. Assess the titration of terlipressin based on mean arterial pressure.
RA6.18. Evaluate the use of point of care ultrasonography to assess intravascular volume and guide albumin treatment in HRS-AKI.
RA6.19. Compare the use of balanced crystalloids *vs*. albumin in HRS-AKI.
RA6.20. Explore the role of urinary biomarker-guided management of HRS-AKI.
RA6.21. Define the role of TIPS in the treatment of HRS-AKI.
RA6.22. Identify strategies of non-antibiotic prophylaxis for SBP.
RA6.23. Develop stopping rules for secondary prophylaxis of SBP beyond recompensation.
RA6.24. Define the optimal timing for TIPS placement after an episode of SBP in patients with recurrent or refractory ascites.

Panel 7 – Cirrhosis recompensation

7.1. Recompensation defines the stage of cirrhosis that has transitioned from a decompensated stage to a compensated stage after aetiologic therapy. Therefore, the term only applies to patients with cirrhosis who have developed overt decompensating events **(LoE 5; new; 96% agreement).**
7.2. Recompensation should not be confused with cirrhosis regression, which denotes histological regression from a cirrhotic to a non-cirrhotic liver architecture **(LoE 5; new; 96% agreement).**
7.3. Recompensation is associated with significantly lower risk of death and HCC when compared to patients who remain decompensated **(LoE 1; new; 93% agreement).**
7.4. The definition of recompensation requires fulfilment of ALL of the following criteria (Fig. 2):
  a. Removal, control or suppression of the primary aetiology/aetiologies of cirrhosis **(LoE 2; modified; 97% agreement).**
  b. Resolution of ascites on imaging in the absence of diuretics (unless indicated for non-ascites-related conditions) and no clinical evidence of HE (off lactulose, rifaximin, LOLA; unless indicated for non-HE-related conditions) **(LoE 2; modified; 93% agreement).**
  c. After variceal bleeding as a decompensating event, absence of recurrent variceal bleeding, regardless of ongoing therapy with carvedilol/cNSBBs **(LoE 2; modified; 95% agreement).**
  d. Criteria (a-c) must be fulfilled for >6 consecutive months to define a state of recompensated cirrhosis **(LoE 4; new; 88% agreement).**
  e. Improvement in liver function tests to a range consistent with a Child-Pugh score of A5/A6 **(LoE 2; modified; 88% agreement).**
7.5. Evidence of recompensation and definitions for aetiologic cure/control exist for the following liver disease aetiologies:
  a. ALD: sustained alcohol abstinence **(LoE 1; new; 94% agreement).**
  b. HBV infection: sustained suppression of HBV-DNA **(LoE 1; new; 100% agreement).**
  c. HCV infection: sustained virological response **(LoE 1; new; 100% agreement).**
7.6. Resolution of ascites while on diuretics is not considered evidence of recompensation **(LoE 5; modified).** Withdrawal of diuretics should be considered in patients who achieve aetiologic cure/control, once ascites has resolved and liver synthetic function has improved to Child-Pugh A laboratory criteria **(LoE 5; strong recommendation; new; 88% agreement).**
7.7. Absence of HE while on specific HE medication (lactulose/rifaximin/LOLA) is not considered evidence of recompensation **(LoE 5; modified).** Withdrawal of HE medication should be considered in patients who achieve aetiologic cure/control once HE has resolved and liver synthetic function has improved to Child-Pugh A laboratory criteria **(LoE 5; strong recommendation; new; 89% agreement).**
7.8. Patients with TIPS may also achieve recompensation, provided all criteria listed in statement 7.4 are fulfilled. However, resolution of ascites *alone* or lack of recurrent variceal bleeding *alone* after TIPS are not considered evidence of recompensation **(LoE 4; modified; 87% agreement).**

7.9. Severity of liver disease at the time of decompensation is a negative predictor of recompensation, with higher Child-Pugh and MELD scores and/or the presence of further decompensation indicating a lower likelihood of recompensation **(LoE 2; new; 94% agreement).**

7.10. Severity of portal hypertension at the time of decompensation seems to be a negative predictor of recompensation, with the presence of high-risk varices indicating a lower likelihood of recompensation **(LoE 3; new; 96% agreement).**

7.11. A shorter time from decompensation to aetiological therapy may increase the potential for recompensation **(LoE 2; new; 90% agreement).**

7.12. The type and number of decompensating events may impact the likelihood of recompensation **(LoE 4; new; 91% agreement).**

7.13. Patients with recompensated cirrhosis remain at risk of liver-related events and should remain under follow-up by a hepatologist at 6-month intervals **(LoE 3; strong recommendation; new; 86% agreement).**

7.14. Even though the risk of HCC in recompensated patients is lower than in decompensated patients, they are still at risk of developing HCC and should thus continue to undergo regular HCC surveillance **(LoE 2; strong recommendation; new; 99% agreement).**

7.15. In patients achieving recompensation, CSPH may resolve over time **(LoE 2; new; 93% agreement).**

7.16. CSPH resolution in recompensated patients can be confirmed by an HVPG <10 mmHg when measured off carvedilol/cNSBBs **(LoE 2; new; 93% agreement).**

7.17. Portosystemic collaterals and varices may persist after recompensation, despite resolution of CSPH. Therefore, they are not reliable indicators of CSPH persistence after recompensation **(LoE 3).** The risk of bleeding from persisting varices after recompensation seems to be low **(LoE 2; new; 96% agreement).**

7.18. Carvedilol/cNSBB treatment may be discontinued in recompensated patients only if resolution of CSPH can be confirmed **(LoE 3; weak recommendation; new; 93% agreement).**

7.19. In patients with recompensated cirrhosis, CSPH resolution/persistence may be non-invasively assessed via LSM and SSM:

a. LSM <10 kPa may rule-out CSPH (LoE 2; weak recommendation; new; 90% agreement).

b. LSM <15 kPa plus SSM <25 kPa may rule-out CSPH **(LoE 4; weak recommendation; new; 92% agreement).**

c. LSM >25 kPa may rule-in CSPH **(LoE 4; weak recommendation; new; 94% agreement).**

Panel 7 – Research agenda

RA7.1. Clearly define "adequate control of metabolic comorbidities" to standardise a surrogate for aetiologic control and facilitate research on cirrhosis recompensation in patients with MASLD and MetALD.

RA7.2. Generate robust evidence supporting the occurrence of recompensation and to define removal or suppression of the primary aetiologic factor(s) in patients with primary biliary cholangitis, autoimmune hepatitis, HDV infection, and hereditary liver diseases (*e.g.* hemochromatosis and Wilson's disease).

RA7.3. Investigate whether differences in length of follow-up and/or cohort characteristics (particularly liver disease severity) may explain the wide range of recompensation rates reported in the literature.

RA7.4. Investigate predictors of cirrhosis recompensation, including structural or other abnormalities present at index decompensation that may define a point of no return and features that resolve or persist among patients who achieve recompensation.

RA7.5. Assess whether an increase in serum albumin levels alone and/or a decrease in MELD score – and their optimal cut-off values are sufficient to define recompensation. Investigate the durability of these laboratory improvements to better characterise recompensation and assess whether sustained improvement provides incremental prognostic value.

RA7.6. Investigate whether baseline hepatic venous pressure gradient (HVPG), LSM, or other noninvasive tests, when performed during decompensation, can predict subsequent recompensation (or lack thereof).

RA7.7. Assess whether the presence, severity, and number of metabolic comorbidities (*e.g.* overweight/obesity or diabetes) impact the likelihood of achieving recompensation and the risk of liver-related events or HCC after aetiologic therapy or cure.

RA7.8. Assess whether dynamics soon after initiation of aetiologic therapy such as a trend towards improvement to CTP-A, increases in serum albumin, and/or improvements in LSM and SSM can predict subsequent recompensation.

RA7.9. Investigate the ability of NITs, including LSM, SSM, blood-based markers, and/or combined models to monitor resolution or persistence of CSPH in recompensated cirrhosis.

RA7.10. Obtain aetiology-specific data to determine the durability of recompensation and the residual risk of decompensation, HCC, and liver-related and all-cause mortality.

RA7.11. Investigate whether pharmacologic agents (*e.g.* statins, FXR agonists, anti-fibrotic or anti-inflammatory drugs) or other therapeutic strategies can facilitate the achievement of recompensation in patients with decompensated cirrhosis.

RA7.12. Investigate the impact of body composition, nutritional support, physical therapy and rehabilitation on the likelihood of achieving and maintaining cirrhosis recompensation.

Panel 8 – Vascular liver diseases, Budd-Chiari syndrome, portosinusoidal vascular disorder, non-cirrhotic portal fibrosis

General considerations in vascular liver disease

8.1. Patients with vascular liver disease (VLD) should be managed in close collaboration with centres that have experience in VLD, particularly when endovascular treatments are being considered **(LoE 4; strong recommendation; new; 96% agreement).**

8.2. In patients with VLD, vaccinations (including viral hepatitis and SARS-CoV-2) and a lifestyle including regular physical activity, a healthy diet and alcohol abstinence should be encouraged. Psychological and social aspects are commonly overlooked in clinical care and require systematic assessment

and management. **(LoE 3; strong recommendation; new; 90% agreement).**

8.3. In adults without cirrhosis, VLDs are frequently associated with ≥1 aetiological factor(s), which may be occult at presentation and should be systematically investigated with a comprehensive workup (detailed information provided in EASL CPG on Vascular Liver Diseases) **(LoE 2; strong recommendation; modified; 97% agreement).**

Anticoagulation in VLD

8.4. Current evidence does not support selecting anticoagulant therapy based on the type of VLD **(LoE 4; new; 98% agreement).**

8.5. When anticoagulation is indicated, therapy can be initiated with low-molecular-weight heparin (LMWH) – followed by the transition to an oral anticoagulant **(LoE 3; weak recommendation).** Because of the increased risk of heparin-induced thrombocytopenia, the use of unfractionated heparin is generally not recommended and should be reserved for special situations (*e.g.* glomerular filtration rate <30 ml/min or pending invasive procedures) **(LoE 3; weak recommendation; unchanged; 96% agreement).**

8.6. When anticoagulation is indicated, direct oral anticoagulants (DOACs) may be the primary choice for oral anticoagulation in the absence of double/triple positive anti-phospholipid syndrome, pregnancy, severely impaired liver function (equivalent to Child-Pugh class C) or impaired renal function (glomerular filtration rate <30 ml/min) **(LoE 3; weak recommendation; modified; 91% agreement).**

8.7. Anticoagulation therapy should be revisited regularly in a multidisciplinary team and adapted according to liver and kidney function, and patient adherence **(LoE 3; strong recommendation; new; 96% agreement).**

Management of portal hypertension-related complications in VLD

8.8. There is currently no evidence to favour EVL over NSBBs for primary prophylaxis of variceal bleeding in patients with VLD **(LoE 5; new; 95% agreement).**

8.9. In VLD, current evidence is insufficient to support the preferential use of carvedilol over other cNSBBs **(LoE 5; new; 95% agreement).**

8.10. When anticoagulation is indicated, it should be initiated without delay and should not be postponed pending endoscopic prophylaxis for variceal bleeding **(LoE 3; strong recommendation; new; 92% agreement).**

8.11. EVL may be safely performed without discontinuing LMWH or vitamin K antagonists (VKAs), while data are lacking regarding DOACs in this setting **(LoE 3; weak recommendation; modified; 92% agreement).**

8.12. In patients with VLD and acute portal hypertension-related bleeding, initial bleeding management and secondary prophylaxis should follow guidelines for patients with cirrhosis **(LoE 4; strong recommendation; modified; 98% agreement).**

8.13. TIPS should be considered in patients with refractory or recurrent portal hypertensive bleeding **(LoE 3; strong recommendation; modified; 98% agreement).**

8.14. There is no data to support the use of pTIPS in VLD **(LoE 5; new; 98% agreement).**

Contraception and pregnancy

8.15. Early counselling should be provided to all women of childbearing age with VLD. This should include guidance on non-oestrogen contraceptive options, discussion of the potential for successful pregnancy under the care of a multidisciplinary team experienced in VLD, and detailed counselling on associated maternal and fetal risks **(LoE 2; strong recommendation; new; 91% agreement).**

8.16. All teratogenic medications, including cytoreductive agents should be discontinued before conception. VKA and DOACs should be discontinued as soon as the pregnancy is detected. Oral anticoagulants should be replaced with therapeutic doses of LMWH during pregnancy **(LoE 2; strong recommendation; new; 93% agreement).**

8.17. In patients with VLD, screening for gastro-oesophageal varices should be performed during the second trimester of pregnancy, unless adequate screening was conducted within the year preceding conception **(LoE 4; strong recommendation; new; 86% agreement).**

8.18. Primary and secondary prophylaxis of variceal haemorrhage during pregnancy should follow the same principles as in non-pregnant patients, including the use of propranolol **(LoE 3; strong recommendation; new; 91% agreement).**

8.19. In the absence of obstetric contraindications, vaginal delivery is generally preferred over caesarean section in patients with portal hypertension, provided the platelet count is >20 G/L **(LoE 3; weak recommendation; new; 88% agreement).**

8.20. Treatment with NSBBs, heparin and warfarin is safe during breastfeeding, whereas other VKAs and DOACs are contraindicated **(LoE 2; new; 88% agreement).**

Budd-Chiari syndrome

8.21. Budd-Chiari syndrome (BCS) is the consequence of an obstruction to the hepatic venous outflow. Obstruction can be located from the level of the small hepatic veins to the level of the entrance of the inferior vena cava into the right atrium **(LoE 1; unchanged; 96% agreement).**

8.22. BCS is the preferred designation for any hepatic venous outflow tract obstruction **(LoE 5; unchanged; 96% agreement).**

8.23. BCS is considered secondary when the mechanism for venous obstruction is an extrinsic compression or endoluminal neoplastic/parasitic obstruction. Otherwise, BCS is considered primary **(LoE 5; modified; 97% agreement).**

8.24. The presentation and manifestations of BCS are extremely diverse, so the diagnosis should be considered in any patient with acute, acute-on-chronic, or chronic liver disease **(LoE 1; strong recommendation; unchanged; 96% agreement).**

8.25. BCS is diagnosed by demonstrating obstruction of the venous lumen, or by the presence of hepatic vein collaterals associated with the absence of patent hepatic veins **(LoE 1; unchanged; 97% agreement).**

8.26. In patients with radiologically confirmed BCS, a liver biopsy is not necessary to establish the diagnosis **(LoE 2; strong recommendation; modified; 93% agreement).**

8.27. When BCS is suspected, and cross-sectional imaging fails to demonstrate large hepatic vein obstruction, liver biopsy is necessary to confirm BCS with small hepatic vein

involvement **(LoE 2; strong recommendation; modified; 96% agreement).**

8.28. In patients with chronic BCS, hepatic nodules are frequent and most often benign. As HCC may also occur, patients should be monitored with periodic imaging and alpha-fetoprotein (AFP) measurements **(LoE 2; strong recommendation; modified; 93% agreement).**

8.29. A 6-month interval can be proposed for periodic imaging and AFP monitoring, similar to HCC surveillance in cirrhosis. An AFP level of >15 ng/ml should raise suspicion for HCC **(LoE 3; strong recommendation; modified; 97% agreement).**

8.30. The LIRADS classification cannot be applied in BCS. MRI with hepatobiliary contrast agents is recommended to help distinguish between benign and malignant lesions. For lesions suspected of being HCC, histological confirmation is needed **(LoE 3; strong recommendation; modified; 91% agreement).**

8.31. Treatment of BCS should follow a stepwise approach including anticoagulation, recanalisation by interventional radiology with or without stent placement, TIPS, and LT **(LoE 2; strong recommendation; unchanged; 97% agreement).**

8.32. Therapeutic anticoagulation should be initiated as soon as possible after the diagnosis of BCS and continued long-term, including after LT **(LoE 2; strong recommendation; modified; 95% agreement).**

8.33. Clinicians should actively assess for venous stenoses that are suitable for percutaneous recanalisation and manage them accordingly **(LoE 2; strong recommendation; modified; 95% agreement).**

8.34. In cases where medical management alone is insufficient to improve symptoms and percutaneous recanalisation is not an option or has failed, TIPS is the recommended next step **(LoE 2; strong recommendation; modified; 97% agreement).**

8.35. Improvement in BCS should be assessed as a combination of the following outcomes, evaluated periodically (*e.g.* at 2-weekly intervals): decreasing rate of ascites formation; decreasing serum bilirubin concentration, decreasing serum creatinine concentration, and/or decreasing international normalised ratio when elevated (and/or increasing factor V in patients receiving VKAs) **(LoE 4; strong recommendation; modified; 95% agreement).**

8.36. The currently available prognostic BCS scores should not determine individual patient management alone, but can be used for research purposes **(LoE 3; strong recommendation; modified; 90% agreement).**

8.37. In patients with BCS presenting with acute liver failure, urgent LT should be considered. Salvage TIPS should be performed, if possible, independently of LT listing **(LoE 3; strong recommendation; unchanged; 93% agreement).**

Panel 8 – Research agenda for BCS

RA8.1. Provide a definition of a centre of expertise for VLD.

RA8.2. Define treatment responses, predictors of improvement, and criteria for stepping up the treatment strategy in BCS.

RA8.3. Determine the role of LSM in evaluating outcome and response to treatment in BCS.

RA8.4. Identify strategies to prevent rethrombosis beyond anticoagulant therapy.

RA8.5. Compare long-term outcomes after TIPS *vs*. no TIPS.

RA8.6. Determine the impact of treatment of the underlying disease on prognosis or treatment response in patients with BCS.

RA8.7. Assess the efficacy and safety of long-term anticoagulation for patients with BCS once the underlying cause has been treated.

RA8.8. Develop a radio-pathological framework for the characterisation of BCS-associated hepatic nodules.

RA8.9. Determine the incidence of HCC, its risk factors and the impact of 6-month surveillance on the development of HCC or hepatic nodules in patients with BCS.

RA8.10. Determine outcomes of HCC in patients with BCS and assess the applicability of traditional HCC LT listing criteria (*e.g.* Milan, AFP model, etc) in LT for HCC in patients with BCS.

RA8.11. Address geographical differences in BCS.

RA8.12. Determine the role of LSM for the early diagnosis of TIPS dysfunction and/or for identifying non-responders to medical treatment.

Portosinusoidal vascular disorder or non-cirrhotic portal fibrosis

8.38. 'Porto-sinusoidal vascular disorder' (PSVD) or 'non-cirrhotic portal fibrosis' (NCPF) represent a broad clinico-pathological entity that can lead to portal hypertension in the absence of cirrhosis, which overlaps with idiopathic portal hypertension or non-cirrhotic intrahepatic portal hypertension and encompasses various histological lesions (see statement 8.43) **(LoE 1; new; 98% agreement).**

8.39. Following the 2026 multi-society nomenclature agreement, both terms 'PSVD' and 'NCPF' are accepted, and they share the same definition (statements 8.38 and 8.40) and diagnostic criteria (statements 8.42, 8.43 and 8.44) **(LoE 5; new; 96% agreement).**

8.40. The absence of portal hypertension does not rule-out PSVD or NCPF. The presence of concomitant causes of cirrhosis (*e.g.* viral hepatitis, excessive alcohol consumption, metabolic syndrome, *etc.*) does not rule-out PSVD or NCPF, and both can coexist. The presence of PVT does not rule out PSVD or NCPF, and both can coexist. (see Table S2 summarising the conditions excluded from the definition of PSVD or NCPF). **(LoE 2; new; 93% agreement).**

8.41. A liver biopsy specimen of ≥15 mm with at least one fragment ≥10 mm – or otherwise considered adequate for interpretation by an expert pathologist – is required for the diagnosis of PSVD or NCPF **(LoE 3; new; 94% agreement).**

8.42. Diagnosis of PSVD or NCPF is based on a scoring system (Fig. 4) that incorporates clinical features of portal hypertension, histological findings, and relevant concomitant conditions. Diagnosis requires ruling-out cirrhosis on an adequate liver biopsy and confirmation that no exclusion criteria are present **(LoE 5; new; 91% agreement).**

8.43. Histological PSVD or NCPF lesions are categorised by major and minor criteria. Major histological criteria include: nodular regenerative hyperplasia (NRH), muscularized portal venules, and portal venule stenosis involving 50% or more of portal tracts. Minor histological criteria include: abnormal distribution of vascular structures, or portal venule stenosis involving 25–49% of portal tracts, or regenerative changes

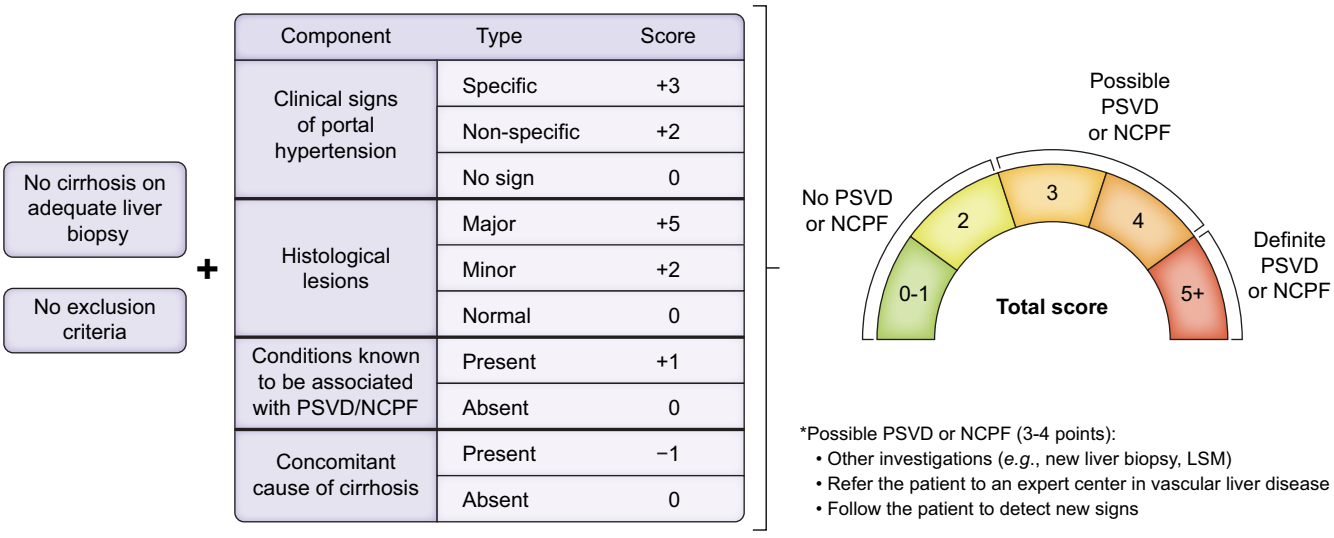


Fig. 4. Diagnostic score for PSVD or NCPF. NCPF, non-cirrhotic portal fibrosis; PSVD, portosinusoidal vascular disorder.

without clear NRH. The concomitant presence of all three minor criteria is considered a major criterion **(LoE 3; new; 84% agreement).**

8.44. Specific signs of portal hypertension include: medium/large oesophageal varices, gastric varices, spontaneous portosystemic collaterals, and SSM >40 kPa in the absence of myeloproliferative neoplasm. Non-specific features of portal hypertension include: clinically overt ascites, small oesophageal varices, thrombocytopenia with platelet counts <150 x 10^9/L, and splenomegaly **(LoE 3; new; 92% agreement).**

8.45. Once the diagnosis of PSVD or NCPF is made, patients should be screened for associated immune/inflammatory/systemic diseases, haematological disorders or genetic disorders and exposure to specific drugs/toxins (see Table S3) **(LoE 4; strong recommendation; modified; 93% agreement).**

8.46. Screening for gastroesophageal varices is recommended at diagnosis of PSVD or NCPF **(LoE 3; strong recommendation).** The recommended frequency of endoscopic surveillance for varices is 2 years in case of no varices, and 1 year in case of small varices **(LoE 3; strong recommendation; modified; 95% agreement).**

8.47. Both at initial diagnosis and during follow-up, endoscopy is not required if SSM is <40 kPa and serum bilirubin is <1 mg/dL (<17 μmol/L), given the low likelihood of detecting varices **(LoE 3; strong recommendation; new; 91% agreement).**

8.48. LSM-based criteria for screening of oesophageal varices used in patients with cirrhosis should not be applied to patients with PSVD or NCPF **(LoE 2; strong recommendation; new; 95% agreement).**

8.49. Contrast-enhanced cross-sectional imaging is recommended at diagnosis of PSVD or NCPF to evaluate the anatomy and patency of the portal venous system, assess for the presence of spontaneous portosystemic collaterals or hepatic veno-venous communications and the presence of nodules **(LoE 4; strong recommendation; new; 96% agreement).**

8.50. Doppler ultrasound every 6 months is recommended for PVT screening in patients with PSVD or NCPF and signs of portal hypertension. The presence of liver nodules may also be assessed during these evaluations **(LoE 3; strong recommendation; new; 95% agreement).**

8.51. There is no data to guide screening for PVT in patients with PSVD or NCPF without features of portal hypertension **(LoE 5; new; 96% agreement).**

8.52. Nodules develop less frequently in PSVD or NCPF than in BCS. Most are benign, but in case of suspicion of HCC, histological confirmation is recommended **(LoE 4; strong recommendation; new; 94% agreement).**

8.53. LT should be considered in patients with PSVD or NCPF and severe and refractory complications of portal hypertension, hepatopulmonary syndrome, HCC, or with advanced liver dysfunction. Indications should be discussed in LT centres with expertise in VLD **(LoE 4; strong recommendation; new; 96% agreement).**

Panel 8 – Research agenda for PSVD or NCPF

RA8.13. Determine the impact of severe associated comorbidities on overall outcomes, liver-related outcomes, and treatment responses in patients with PSVD or NCPF.

RA8.14. Characterise the natural history of PSVD or NCPF in the absence of portal hypertension.

RA8.15. Determine the incidence, prognosis and predictors of PVT development in patients with PSVD or NCPF, and the impact of anticoagulation.

RA8.16. Identify strategies for PVT prophylaxis in patients with PSVD or NCPF and signs of portal hypertension.

RA8.17. Develop and validate non-invasive methods for diagnosis, prognosis, or treatment response (*e.g.* cross-sectional imaging, SSM) in patients with PSVD or NCPF.

RA8.18. Determine the diagnostic and prognostic value of EUS-guided portal pressure gradient measurement in patients with PSVD or NCPF.

RA8.19. Assess the benefit of carvedilol over cNSBBs in patients with PSVD or NCPF.
RA8.20. Develop a radio-pathological framework for the characterisation PSVD or NCPF-associated hepatic nodules.
RA8.21. Histological progression or regression of PSVD- or NCPF-related lesions.
RA8.22. Characterise paediatric PSVD or NCPF.
RA8.23. Determine the impact of PSVD or NCPF on quality of life.
RA8.24. Determine the role of EUS-guided portal pressure gradient measurement and liver biopsy in the management of patients with PSVD or NCPF.

Panel 9 – VALDIG 2 - portal vein thrombosis

9.1. Portal vein thrombosis (PVT) is characterised by the presence of a thrombus in the portal vein trunk or its branches. In the context of PVT, portal cavernoma corresponds to the presence of multiple dilated peribiliary and/or gallbladder wall veins (≥2 mm) irrespective of the patency of the main portal vein that develop as a consequence of prior portal vein obstruction **(LoE 5).** Recent PVT is defined by the formation of a thrombus within the last 6 months. Acute PVT refers to symptomatic recent PVT. Chronic PVT refers to persistent portal vein obstruction lasting more than 6 months, with or without cavernoma **(LoE 2; modified; 90% agreement).**

9.2. Doppler ultrasound is the first-line diagnostic tool, showing solid intraluminal material, suggesting PVT, and/or a network of porto-portal collaterals, suggesting cavernoma. Multiphase CT should be performed to confirm PVT diagnosis, assess local factors, determine the extent of PVT, evaluate involvement of other splanchnic veins, and identify complications and signs of portal hypertension. Contrast-enhanced MRI is an alternative **(LoE 2; strong recommendation; modified; 92% agreement).**

9.3. According to the VALDIG PVT criteria, the degree of occlusion should be assessed as the percentage of remnant portal lumen (RL) compared to its largest diameter. The degree of occlusion should be classified as follows: minimal if RL ≥50%; partial if RL <50%; complete if RL = 0%, by CT/MRI in the portal venous phase, with the plane strictly perpendicular to the main portal vein **(LoE 4; strong recommendation; modified; 93% agreement).**

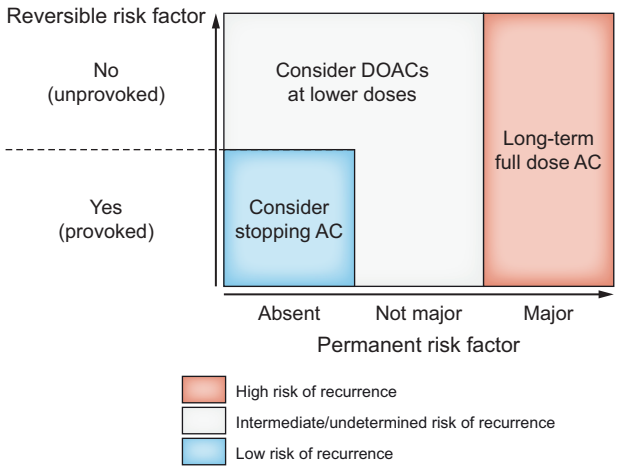


Fig. 5. Considerations for long-term anticoagulation for PVT in the absence of cirrhosis. Recommendation for anticoagulation for PVT in the absence of cirrhosis. In patients without underlying liver disease the risk of recurrence of thrombosis and indication for long-term anticoagulation is based on the absence/presence of a permanent major or non-major risk factor and on the classification of PVT as provoked or unprovoked. Blue boxes refer to patients with low risk of recurrence of thrombosis, red boxes refer to cases with high-risk of recurrence of thrombosis, and grey boxes refer to patients in whom the risk of recurrence of thrombosis is either intermediate or undetermined. PVT is considered provoked if it develops in the setting of a recent (<3 months) reversible risk factor. In all other situations, PVT is considered unprovoked. PVT is considered idiopathic only after exclusion of all permanent and reversible risk factors listed in Table S4. AC, anticoagulation; DOACs, direct oral anticoagulants; PVT, portal vein thrombosis.

PVT in the absence of cirrhosis

Aetiological workup

9.4. Cirrhosis/cACLD should be investigated by a comprehensive workup, including LSM. Liver biopsy and HVPG measurement should be considered in inconclusive cases or when PSVD or NCPF is suspected **(LoE 4; strong recommendation; modified; 93% agreement).**

9.5. Aetiological workup should include screening for permanent and reversible risk factors (see Table S4). Permanent risk factors can be further classified into major (including thrombophilia and myeloproliferative neoplasm) and non-major risk factors, depending on the risk of thrombosis recurrence **(LoE 3; strong recommendation; new; 89% agreement).**

Anticoagulation

9.6. Recent PVT rarely resolves spontaneously, and recanalisation correlates with early initiation of anticoagulation. Therefore, anticoagulation should be started immediately at diagnosis with therapeutic dosing **(LoE 2; strong recommendation; unchanged; 92% agreement).**

9.7. In patients with PVT and recent extension to the superior mesenteric vein (SMV), clinical and radiological signs of intestinal ischaemia should be assessed **(LoE 3; strong recommendation; modified; 97% agreement).**

9.8. In patients with recent PVT, at least 6 months of anticoagulation is recommended to prevent extension and achieve recanalisation **(LoE 2; strong recommendation; unchanged).** However, discontinuation of anticoagulation may be considered in case of complete recanalisation of the portal vein after 3 months, in patients with complete resolution of a reversible risk factor and in the absence of a permanent risk factor **(LoE 5; weak recommendation; new; 94% agreement).**

9.9. In patients with recent PVT extending into the SMV and/or splenic vein who fail to achieve recanalisation after 6 months, full dose anticoagulation therapy may be continued for an additional 6 months, as delayed recanalisation can still occur **(LoE 4; weak recommendation; modified; 93% agreement).**

9.10. The indication for long-term anticoagulation depends on the type of risk factor (see Fig. 5):

a. Long-term anticoagulation at therapeutic doses is recommended in patients with a permanent major risk factor for thrombosis **(LoE 3; strong recommendation; modified; 100% agreement).**

b. In patients without a permanent major risk factor for thrombosis, long-term DOACs at lower dose can be considered to prevent thrombosis recurrence **(LoE 2; weak recommendation; modified; 100% agreement).**
c. After complete resolution of a reversible risk factor, discontinuation of anticoagulation can be considered in the absence of a permanent risk factor **(LoE 4; weak recommendation; modified; 100% agreement).**

Management of portal hypertension

9.11. Upper GI endoscopy should be performed at diagnosis. In patients without high-risk varices, upper GI endoscopy should be repeated at 12 months and every 2 years thereafter. In patients with SSM <40 kPa, endoscopy can be spared because the risk of high-risk varices is low **(LoE 3; strong recommendation; modified; 89% agreement).**

Interventional radiology

9.12. Endovascular restoration of portal vein flow relies on a range of techniques, including thrombolysis and thrombectomy for acute/recent PVT, and angioplasty (with/without stenting) and/or TIPS placement for chronic PVT. Technical feasibility and treatment selection depend on the age and extent of thrombosis and local expertise **(LoE 5; strong recommendation; new; 93% agreement).**

9.13. Endovascular restoration of portal vein flow includes TIPS placement in patients with increased intrahepatic resistance (PSVD, PVT extension to the intrahepatic branches) **(LoE 5; weak recommendation; new; 90% agreement).**

9.14. In patients with PVT extending to the SMV, if bowel ischaemia persists or worsens after 24–48 h of anticoagulation (by clinical reassessment, blood tests including lactate levels, and repeated CT scan) and the patient does not require immediate intestinal surgical resection, endovascular treatment should be considered in experienced centres to prevent/reduce the need for surgical intestinal resection **(LoE 4; strong recommendation; modified; 93% agreement).**

9.15. In patients with chronic PVT, with or without cavernoma, and in the absence of cirrhosis, portal vein flow restoration should be considered in those with recurrent complications of PH or symptomatic portal cavernoma cholangiopathy **(LoE 4; strong recommendation; modified; 96% agreement).**

Panel 9 - Research agenda for PVT without cirrhosis

RA9.1. Understand the effect of aetiological treatment of underlying liver disease on PVT risk and outcomes.
RA9.2. Evaluate the efficacy of long-term low dose anticoagulation in preventing recurrence of thrombosis in patients with a major risk factor.
RA9.3. Assess the efficacy of DOACs (other than rivaroxaban 15 mg/day) in patients with chronic PVT and no major risk factor for thrombosis.
RA9.4. Assess the efficacy and the safety of anticoagulation interruption in patients with low plasma D-dimer concentration after resolution of a transient risk factor for PVT and no persistent risk factor for thrombosis.
RA9.5. Assess the efficacy and safety of long-term anticoagulation in patients with elevated plasma factor VIII concentrations and high-risk molecular variants detected by NGS who have no major risk factors for PVT.
RA9.6. Assess the safety of EVL in patients treated with DOACs.
RA9.7. Define the optimal timing and patient selection criteria for endovascular restoration of portal vein flow in recent or acute PVT without bowel ischaemia, particularly in patients who do not achieve recanalisation after anticoagulation or who have a low likelihood of spontaneous recanalisation.
RA9.8. Evaluate the beneficial impact of earlier endovascular intervention on long-term outcomes in prospective studies.

PVT in patients with cirrhosis

Screening

9.16. Screening for PVT is recommended in patients with cirrhosis at the same time as radiological surveillance for HCC **(LoE 3; strong recommendation; modified; 96% agreement).**

9.17. Occurrence of PVT in the presence of HCC is not proof of vascular invasion. Contrast-enhanced imaging is required, and use of standardised criteria for assessment is recommended (*e.g.* A-VENA, LIRADS) **(LoE 3; strong recommendation; modified; 97% agreement).**

Indications for treatment

9.18. In potential LT candidates, the goal of anticoagulation is to prevent re-thrombosis or progression of thrombosis to facilitate porto-portal anastomosis and reduce post-transplant morbidity and mortality. Anticoagulation is recommended in potential LT candidates with any PVT **(LoE 3; strong recommendation; modified; 93% agreement).**

9.19. Anticoagulation should be considered in patients with (i) recent partial (<50% remnant lumen) or complete thrombosis of the main portal vein **(LoE 3; strong recommendation),** (ii) symptomatic PVT, independently of the extension **(LoE 3; strong recommendation),** (iii) nontumoral PVT and HCC, especially in those with an indication for locoregional therapy **(LoE 3; weak recommendation; modified; 90% agreement).**

9.20. In candidates for LT, anticoagulation should be continued until surgery; in all other patients, it should be maintained for at least 6 months and considered for long-term use, regardless of recanalisation **(LoE 3; strong recommendation; modified; 97% agreement).**

9.21. If anticoagulation is discontinued, close follow-up with imaging at 1 month, and then at 3 months intervals thereafter is recommended to evaluate recurrence/progression of thrombosis **(LoE 5; strong recommendation; new; 97% agreement).**

9.22. In patients with cirrhosis and PVT who are not treated with anticoagulation, repeated imaging is recommended within 6 weeks of diagnosis to assess for progression of PVT. For patients who do not develop progression, close follow-up with imaging at 3-month intervals is recommended, particularly in patients listed for LT. For patients who develop progression during follow-up, anticoagulation is indicated **(LoE 5; weak recommendations; new; 94% agreement).**

Considerations for anticoagulant therapy in patients with cirrhosis

9.23. Certain characteristics may place patients at higher risk of bleeding complications with anticoagulant therapy, including age, frailty, renal insufficiency (*e.g.* glomerular filtration rate below 30 ml/min), decompensated liver disease, and severe thrombocytopenia (*e.g.* PLT <50 G/L). Decisions regarding anticoagulation should be individualised based on the expected benefit–risk balance. LMWH, VKAs, and DOACs are accepted anticoagulation therapies for patients with cirrhosis and PVT. Benefit–risk balance should be reassessed routinely for change in risk **(LoE 3; strong recommendation; modified; 91% agreement).**

9.24. DOACs are safe in patients with Child-Pugh class A cirrhosis and may be associated with reduced bleeding risk compared to VKAs. DOACs should be used with caution in patients with impaired liver function (equivalent to Child-Pugh class B). The use of DOACs in patients with severe liver dysfunction (equivalent to Child-Pugh C) is not recommended outside of research protocols **(LoE 3; strong recommendation; modified; 97% agreement).**

9.25. DOACs may have different safety-efficacy profiles in patients with cirrhosis. Currently no recommendation can be made on the choice of a specific DOAC. **(LoE 3; weak recommendation; modified; 96% agreement).**

Interventional radiology

9.26. In patients with cirrhosis and PVT, endovascular restoration of portal vein flow using TIPS can be considered in those with main portal vein thrombosis that does not improve during anticoagulation therapy and who either (i) have complications of portal hypertension or (ii) are potential LT candidates, with the aim of facilitating physiological portal vein anastomosis during LT **(LoE 3; strong recommendation; modified; 97% agreement).**

9.27. TIPS may be considered as a first-line treatment in patients with main portal vein thrombosis with complications of portal hypertension and contraindications to anticoagulation therapy **(LoE 4; weak recommendation; new; 93% agreement).**

Table 3. Statements of previous Baveno consensus reports that remain valid.

Statements on medical management	
The decision to treat with Carvedilol/cNSBB should be taken when indicated, independent of the possibility of measuring HVPG	Baveno VI, VII
Alcohol abstinence should be encouraged in all patients with cirrhosis irrespective of aetiology	Baveno VI
The use of aspirin should not be discouraged in patients with cirrhosis and an approved indication for aspirin, since it may reduce the risk of HCC, liver-related complications, and death	Baveno VII
PHG has to be distinguished from gastric antral vascular ectasia because treatments are different	Baveno VI, VII
Short-term albumin administration is indicated for spontaneous bacterial peritonitis (SBP), acute kidney injury (AKI) >stage 1A, large-volume paracentesis and combined with terlipressin for hepatorenal syndrome (HRS)-AKI	Baveno VII
TIPS is the treatment of choice in patients who rebleed despite traditional NSBBs or carvedilol and EVL	Baveno VI, VII
In patients who bleed despite adherence to traditional NSBBs or carvedilol as primary prophylaxis, the combination of traditional NSBBs or carvedilol and EVL is recommended, and TIPS should be considered in those with recurrent ascites	Baveno VII
In all patients hospitalized with decompensation, bacterial infections should be ruled out. The minimal work-up for infections should include diagnostic paracentesis, chest-X ray, cultures of blood, ascites and urine and skin examination	Baveno VII
Patients with bacterial infections should be promptly treated with antibiotics. The empirical antibiotic treatment should be tailored to local epidemiology, risk factors for multidrug-resistant bacteria and severity of infection. If no response to antibiotics is observed, consider viral and fungal infections	Baveno VII
Statements on the value of HVPG*	
HVPG measurement provides prognostic information. It is the gold-standard method to assess the presence of CSPH which is defined as HVPG ≥10 mmHg	Baveno VI, Baveno VII
In patients with viral- and alcohol-related cirrhosis, HVPG measurement is the gold-standard method to determine the presence of "clinically significant portal hypertension" (CSPH), which is defined as an HVPG ≥10 mmHg	Baveno VII
In patients with MASLD-related cirrhosis, although an HVPG ≥10 mmHg remains strongly associated with the presence of clinical signs of portal hypertension, these signs can also be present in a small proportion of patients with HVPG values <10 mmHg	Baveno VII
The presence of CSPH, determined either by HVPG ≥10 mmHg or by clinical manifestations of portal hypertension is associated with a higher risk of decompensation and mortality in patients with cirrhosis undergoing liver resection for hepatocellular carcinoma	Baveno VII
In candidates for non-hepatic abdominal surgery, a HVPG ≥16 mmHg is associated with an increased risk of short-term mortality after surgery	Baveno VII
Removal/suppression of the primary aetiological factor leads to potentially meaningful decreases in HVPG in most patients and substantially reduces the risk of hepatic decompensation	Baveno VII
Absence/resolution of CSPH following removal/suppression of the primary aetiological factor prevents hepatic decompensation	Baveno VII
The optimal percent/absolute decrease in HVPG associated with a reduction in hepatic decompensation following the removal/suppression of the primary aetiological factor in patients with cACLD and CSPH has yet to be established	Baveno VII
HVPG change is a relevant surrogate outcome. HVPG response to NSBBs is associated with a significant reduction in risk of variceal bleeding and decompensation	Baveno VI
Measurement of HVPG response to therapy offers additional relevant information: a decrease in HVPG of at least 10% from baseline or to ≤12 mmHg after chronic treatment with NSBB is clinically relevant in the setting of primary prophylaxis. Similarly, acute HVPG response to intravenous propranolol may be used to identify responders to NSBB, specifically a decrease in HVPG of 10% or to ≤12 mmHg may be relevant in this setting	Baveno V, VI
HVPG measurements should be encouraged in clinical trials investigating novel therapies but are not essential if portal hypertension-associated endpoints are well defined	Baveno VI

(*continued on next page*)

Table 3. (*continued*)

Statements on technical aspects of HVPG and PPG measurement	
The use of an end-hole, compliant balloon occlusion catheter reduces the random error of wedged hepatic vein pressure (WHVP) measurements and is preferred over the use of conventional straight catheter	Baveno VII
A small volume of contrast medium should be injected when the occlusion balloon is inflated to confirm a satisfactory occluded position and to exclude the presence of hepatic vein-to-vein communications	Baveno VII
Deep sedation during liver haemodynamic measurement may cause inaccurate HVPG values. If light sedation is required, low dose midazolam (0.02 mg/kg) does not modify the HVPG and is acceptable	Baveno VII
To properly reflect portal venous pressure, WHVP requires a stabilisation time. Recording of WHVP requires a minimum of 1 min, with particular attention to stability during the last 20-30 s. WHVP should be recorded in triplicate	Baveno VII
The wedged to free hepatic vein pressure gradient has superior clinical prognostic value than wedged to right atrial pressure gradient and should be used as the standard reference for HVPG calculation. Right atrial pressure (RAP) can be measured to rule out post-hepatic component of portal hypertension	Baveno VII
The immediate post-TIPS PPG may be influenced by various factors, such as general anaesthesia, use of vasoactive agents or haemodynamic instability and therefore immediate post-TIPS PPG may not represent long-term PPG. PPG measurements in haemodynamically stable, non-sedated patients better reflect post-TIPS PPG values and are recommended	Baveno VII

*These statements remain valid but should be interpreted in the context of current Baveno VIII recommendations which favour non-invasive tests over routine HVPG measurements in daily clinical practice.

9.28. In patients with PVT and cirrhosis who undergo TIPS, routine anticoagulation after TIPS is not recommended **(LoE 2; strong recommendation; new; 96% agreement).**

Panel 9 – Research agenda for PVT in cirrhosis

RA9.9. Assess the VALDIG PVT criteria as surrogates for hard clinical outcomes in patients with cirrhosis and PVT.

RA9.10. Standardise future research definitions for bleeding outcomes in studies assessing anticoagulation and PVT.

RA9.11. Assess the efficacy and the safety of TIPS as a first-line treatment in patients with PVT.

RA9.12. Evaluate the efficacy and safety of DOACs in patients with decompensated cirrhosis.

RA9.13. Assess the safety of variceal band ligation in patients treated with DOACs.

Chronic PVT in children

9.29. Chronic PVT in paediatrics is defined as the complete obstruction to the normal flow of portal vein blood to the liver. It can be congenital or acquired, and it is typically diagnosed on imaging **(LoE 3; new; 96% agreement).**

9.30. Diagnostic work up should include standard evaluations for underlying chronic liver disorders. Liver biopsy may be considered to rule out intrinsic liver disease. Investigation of potential thrombophilia is recommended **(LoE 3; strong recommendation; new; 98% agreement).**

9.31. In the presence of features of portal hypertension, Meso-Rex bypass should be considered for preprimary (*i.e.* without the presence of varices) and primary prophylaxis of variceal haemorrhage and/or other complications of portal hypertension in children. To ensure a >90% success rate, established criteria should be met, which include but are not limited to patency of the intrahepatic portal venous system and surgical expertise in performance of the Meso-Rex bypass **(LoE 3; strong; new; 100% agreement).**

9.32. The role of other types of primary prophylaxis (*i.e.* pharmacologic, endoscopic and interventional radiologic) of variceal haemorrhage is not well established. **(LoE 4; new; 95% agreement).**

9.33. The approach to management of varices may need to be individualised for children without ready access to medical care **(LoE 4; weak recommendation; new; 90% agreement).**

9.34. Meso-Rex bypass is the preferred approach to secondary prophylaxis of variceal haemorrhage in children with chronic PVT. If Meso-Rex bypass is not feasible, portal vein recanalisation, endoscopic treatment with or without NSBBs, or distal splenorenal shunting should be considered **(LoE 3; strong recommendation; new; 100% agreement).**

9.35. Distal splenorenal shunting should be considered for recurrent variceal haemorrhage in the absence of hepatopulmonary syndrome, portopulmonary hypertension and/or HE **(LoE 3; strong recommendation; new; 93% agreement: 93%).**

9.36. In patients with portal hypertension and/or portosytemic shunts (including both spontaneous and interventional), annual surveillance is warranted to monitor for complications related to portosystemic shunting, especially occurrence of portopulmonary hypertension **(LoE 4; strong recommendation; new; 93% agreement: 93%).**

9.37. In the absence of alternative therapeutic approaches, LT should be considered in children with potentially life-threatening and otherwise untreatable portal hypertensive complications **(LoE 3; strong recommendation; new; 100% agreement: 100%)**

Recent PVT and PVT in children with cirrhosis

9.38. Recent PVT has been described in neonates with umbilical catheters. Spontaneous resolution of PVT can occur in neonates and as such the role of anticoagulation in neonates with recent PVT is uncertain. Monitoring for clot extension is warranted. In the presence of an extensive thrombus, anticoagulant therapy may be considered **(LoE 4; weak recommendation; new; 100% agreement: 100%).**

9.39. PVT associated with cirrhosis is uncommon in children. Thus, the evidence supporting the translation from adult recommendations to children with cirrhotic PVT is insufficient **(LoE 5; new; 93% agreement).**

Panel 9 – Research agenda for paediatric PVT

RA9.14. Standardise future research definitions for bleeding outcomes in studies assessing anticoagulation.

RA9.15. Determine the prevalence and clinical relevance of inherited and acquired thrombophilia in children with chronic PVT and assess its impact on recurrence risk and long-term management.

RA9.16. Assess the comparative effectiveness of different strategies for primary and secondary prophylaxis of variceal bleeding in children with chronic PVT, including endoscopic, pharmacological, interventional radiologic and surgical approaches.

RA9.17. Evaluate the feasibility, efficacy, and long-term patency of portal vein recanalisation in children with chronic PVT.

RA9.18. Determine the role and safety of anticoagulation in paediatric recent PVT, particularly in catheter-related thrombosis.

Abbreviations

ACLD, advanced chronic liver disease; ACLF, acute-on-chronic liver failure; AFP, alpha-fetoprotein; AKI, acute kidney injury; ALD, alcohol-related liver disease; AVB, acute variceal bleeding; BCLC, Barcelona Clinic Liver Cancer; BCS, Budd-Chiari syndrome; cACLD, compensated advanced chronic liver disease; cNSBB, conventional non-selective beta-blocker; CSPH, clinically significant portal hypertension; DOAC, direct oral anticoagulant; EASL, European Association for the Study of the Liver; EGD, oesophago-gastro-duodenoscopy; ELF, enhanced liver fibrosis (score); EVL, endoscopic variceal ligation; FD, further decompensation; FIB-4, Fibrosis-4 (index); FMT, faecal microbiota transplant; GOV, gastro-oesophageal varices; HCC, hepatocellular carcinoma; HE, hepatic encephalopathy; HRS-AKI, hepatorenal syndrome-acute kidney injury; HVPG, hepatic venous pressure gradient; IGV, isolated gastric varices; LMWH, low-molecular-weight heparin; LOLA, L-ornithine-L-aspartate; LoE, level of evidence; LSM, liver stiffness measurement; LT, liver transplantation; MASLD, metabolic dysfunction-associated steatotic liver disease; MELD, model for end-stage liver disease; NCPF, non-cirrhotic portal fibrosis; NIT, non-invasive test; NSBB, non-selective beta-blocker; NRH, nodular regenerative hyperplasia; OHE, overt hepatic encephalopathy; OV, oesophageal varices; PHG, portal hypertensive gastropathy; PLT, platelet count; PPIs, proton pump inhibitors; PSVD, portosinusoidal vascular disorder; pTIPS, pre-emptive transjugular intrahepatic portosystemic shunt; PVT, portal vein thrombosis; RCT, randomised controlled trial; SBP, spontaneous bacterial peritonitis; SLD, steatotic liver disease; SMV, superior mesenteric vein; SSM, spleen stiffness measurement; TIPS, transjugular intrahepatic portosystemic shunt; VKA, vitamin K antagonist; VLD, vascular liver disease.

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Supplementary data

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Supplemental information

Baveno VIII – Advancing consensus in portal hypertension

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Table S1. Standardized tools to evaluate frailty and sarcopenia

Frailty	Sarcopenia
Objective performance-based measures	**Objective imaging-based measures**
• **Liver Frailty Index (LFI)*** • Fried Frailty phenotype • Short Physical Performance Battery • Hand Grip Strength • 6-min walk test (6MWT) • Gait speed	• **CT-derived skeletal muscle index (SMI) at L3*** • CT-derived transversal psoas muscle thickness index (TPMT-Index) at L3 • MRI-derived muscle measures
	Objective body composition / bedside measures
	• Dual-energy X-ray absorptiometry (DEXA) • Bioelectrical impedance analysis (BIA) • Ultrasound-based muscle assessment • Anthropometric measures

(*) Most supporting evidence exists for the LFI and the SMI as validated tools for the assessment of frailty and sarcopenia.

Table S2. Conditions excluded from the definition of PSVD or NCPF

Exclusion Criteria for PSVD or NCPF
● Abernethy syndrome ● Budd Chiari syndrome (BCS) or hepatic venous outflow obstruction ● Heart failure or Fontan Associated Liver Disease (FALD) ● Cholestatic liver diseases ● Congenital hepatic fibrosis ● Gaucher disease ● Hepatic amyloidosis ● Hereditary hemorrhagic telangiectasia ● Hypervitaminosis A ● Intrahepatic arteriovenous shunts ● Liver infiltration by tumor cells ● Peliosis hepatis ● Schistosomiasis [(1)] ● SOS in the last 6 months [(2)]

(1) In endemic areas and in the absence of hepatosplenic schistosomiasis, a positive serology or the incidental finding of a few parasite eggs in the liver can occur in PSVD and are not exclusion criteria

(2) Six months or more after bone marrow transplantation, PSVD can develop in patients previously diagnosed with SOS (sinusoidal obstruction syndrome)

Table S3. Drugs/conditions associated with PSVD/NCPF

Category	Condition/Exposure
Drug Exposure	Azathioprine
	6-Thioguanine exposure
	Methotrexate exposure
	Oxaliplatin
	Didanosine
	Stavudine
	Arsenic exposure
	Vinyl chloride
	Trastuzumab-Emtansine
Genetic Diseases	Adams-Oliver Syndrome (*NOTCH1, ARHGAP31*, etc.)
	Telomere Biology Disorders (*TERT, TERC, DKC1*, etc.)
	Retinal Vasculopathy with Cerebral Leukoencephalopathy and Systemic manifestations (*TREX1*)
	Chronic Granulomatous Disease (*CYBB, NCF1, NCF2*, etc.)
	X-Linked Agammaglobulinemia (Bruton's) (*BTK*)
	Activated PI3K Delta Syndrome (*PIK3CD, PIK3R1*)
	Hyper IgM Syndrome (*CD40L, AICDA, UNG*, etc.)
	Cystic Fibrosis (*CFTR*)
	Cystinosis (*CTNS*)
	William-Beuren Syndrome (ELN)
	Turner's Syndrome
	Pierpont Syndrome (*TBL1XR1*)
	Developmental Disorders (*NOTCH1, CTC1*)
	CNN3 gene mutation
	DGUOK gene mutation
	Familial Obliterative Portal Venopathy (*FOPV*)
	GIMAP5 gene mutation
	FCHSD1 gene mutation
	TRMT5 gene mutation
	HRG gene mutation
	TACI gene mutation
	Deficiency of Adenosine Deaminase 2 (*DADA2*)
Hematologic and Prothrombic Diseases	Anti-phospholipid antibodies and anti-phospholipid antibody syndrome
	Protein C deficiency
	Protein S deficiency
	Antithrombin deficiency
	Factor V Leiden mutation
	Prothrombin gene mutation
	ADAMTS13 deficiency
	Hodgkin's lymphoma
	Multiple myeloma
	Marginal B cell lymphoma
	Monoclonal gammopathy of uncertain significance
	Chronic lymphoid leukemia
	Paroxysmal hemoglobinuria
	Aplastic anemia
	Idiopathic thrombocytopenia purpura
	Myeloproliferative neoplasms
Immune/Inflammatory Diseases	Common variable immunodeficiency syndrome
	Castleman disease
	Systemic lupus erythematosus
	Progressive systemic sclerosis
	Rheumatoid arthritis / Felty's syndrome
	Still's disease
	Vasculitis
	Sjögren syndrome
	Mixed connective tissue disease
	Inflammatory bowel disease
	Celiac disease
	POEMS syndrome
	Autoimmune nephropathy
	Behçet disease
	Sarcoidosis without biliary/centrilobular involvement
	Dermatomyositis
Transplantation	Solid organ transplantation
Infection	Recurrent/chronic low-grade abdominal infections
	Human Immunodeficiency Virus

Table S4. Classification of risk factors for PVT in the absence of cirrhosis

Permanent risk factors		Reversible risk factors
Major	**Not major**	
● Myeloproliferative neoplasm ● Nocturnal paroxysmal hemoglobinuria ● Antiphospholipid syndrome ● Behcet's disease ● Homozygous or heterozygous composite thrombophilia gene mutations. ● Congenital antithrombin/ protein C/ protein S deficiency ● Active cancer ● Personal or 1st degree family of history deep vein thrombosis	● Isolated heterozygous G20210A factor II or G1691A factor V mutation ● Hyperhomocysteinemia ● Central obesity ● High risk NGS variants	Local factor: - Abdominal trauma - Abdominal surgery - Abdominal inflammation/ infection Estrogen-containing contraception Pregnancy CMV infection SARS-cov2 infection SARS-cov 2 vaccination